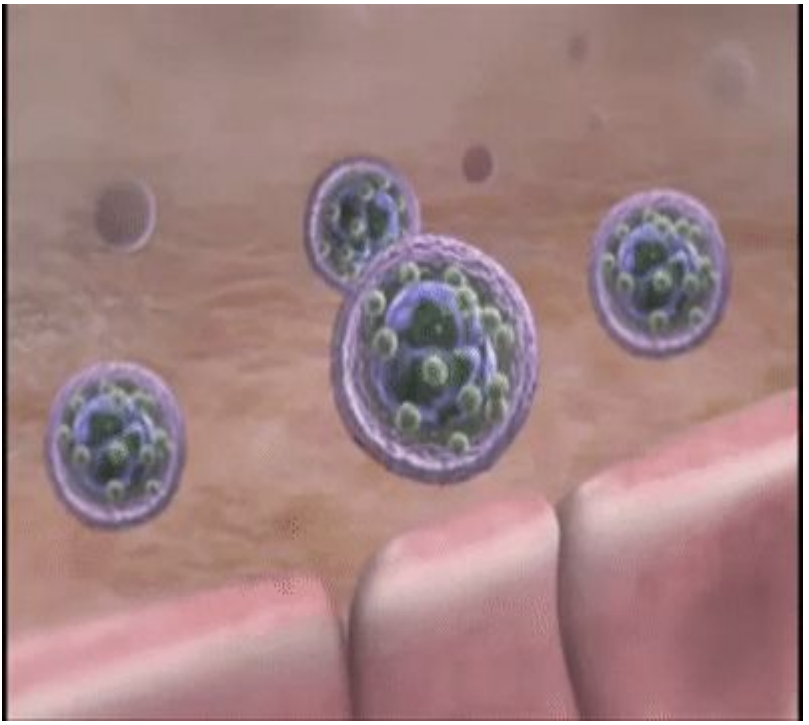
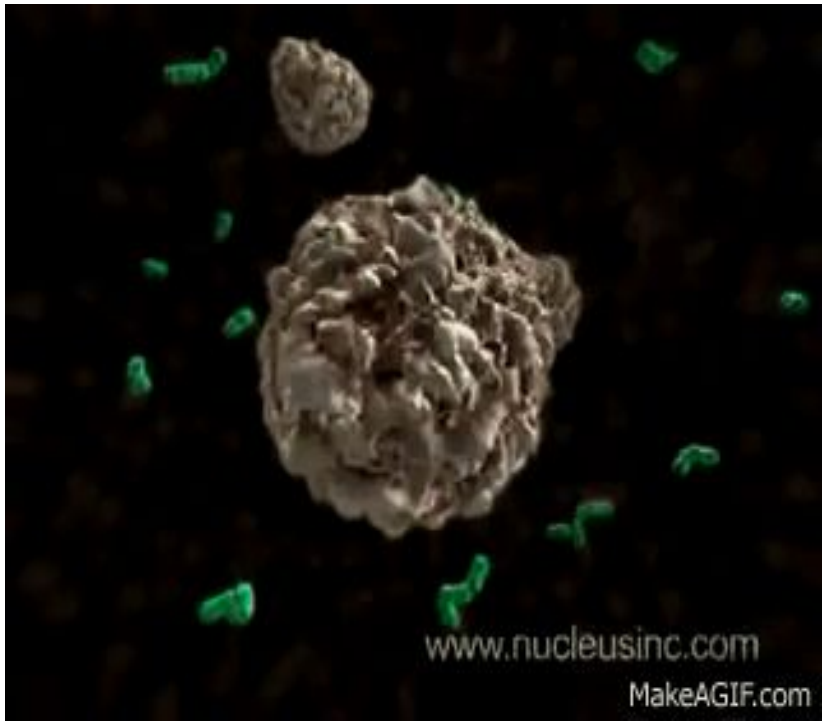
Acute Inflammation
(Mediators and Morphological Patterns)

durre.sameen@duhs.edu.pk





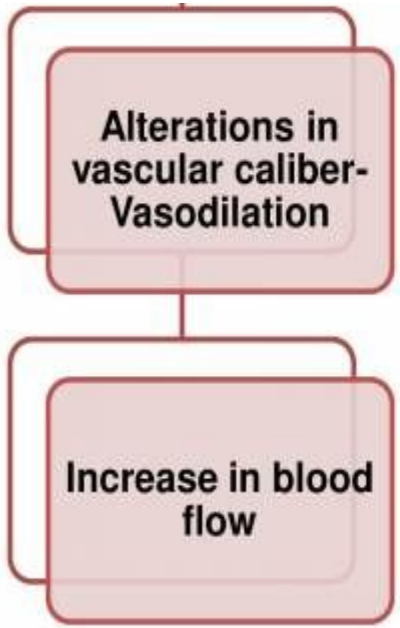


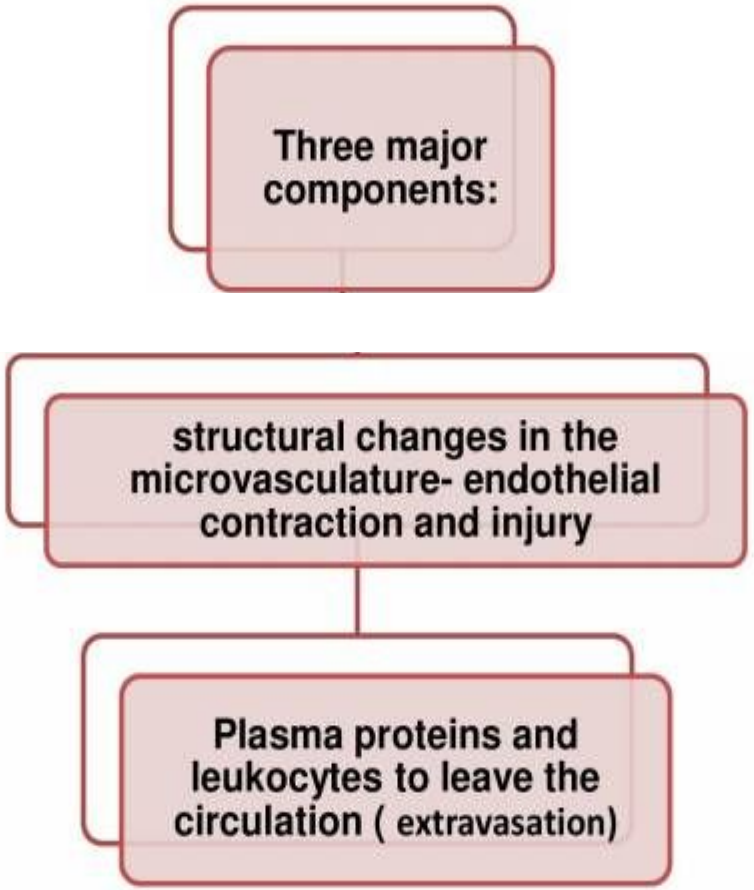
A Quick Recap

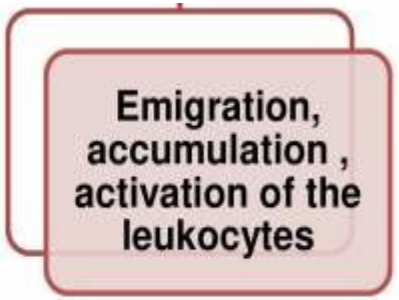


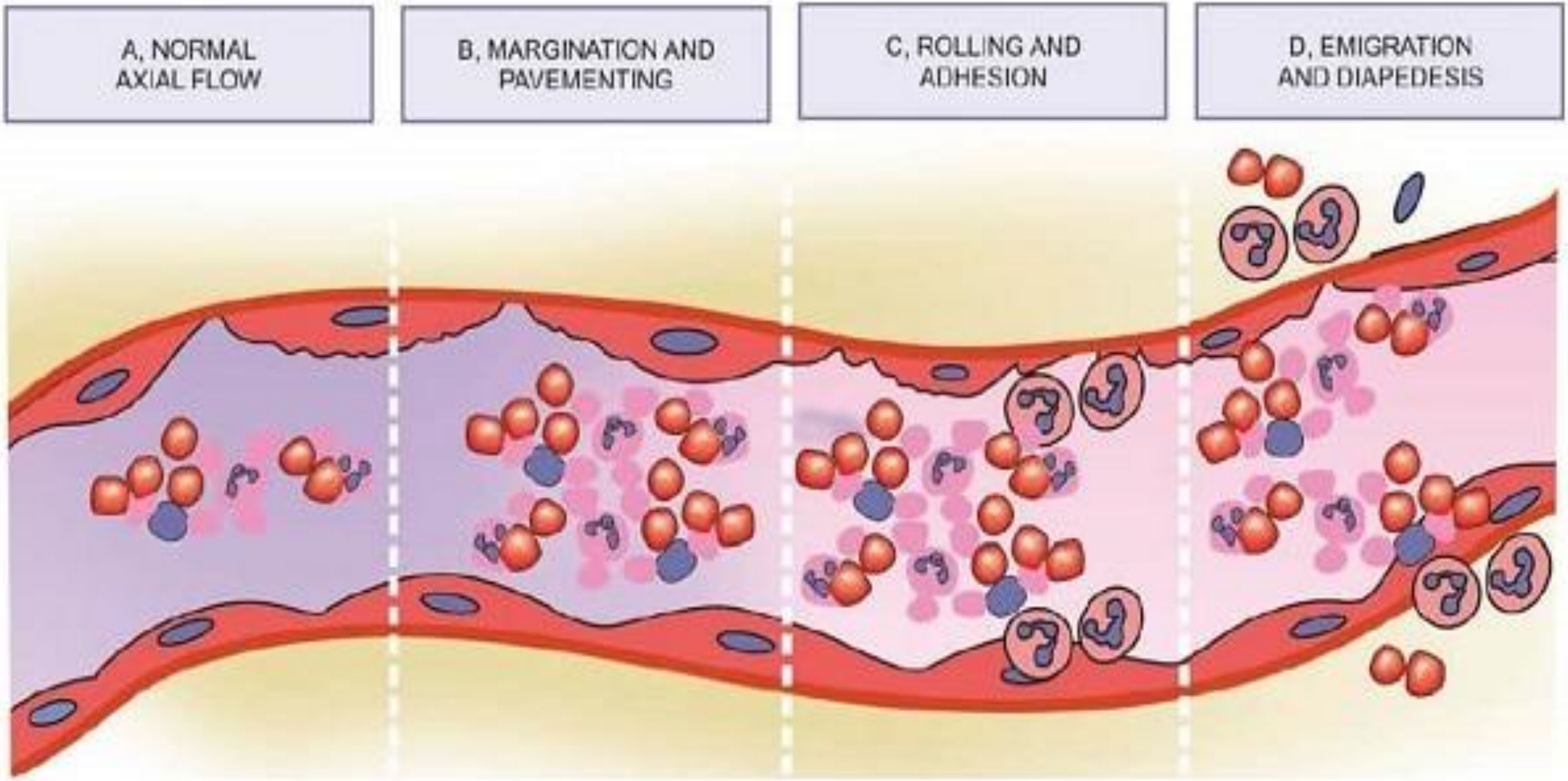
- Name stages of inflammation
- What are three components of acute inflammation
- Name the steps of leukocyte recruitment to sites of Inflammation

- Recognition of injurious agent
- Recruitment of leukocytes
- Removal of the agent
- Regulation of response
- Resolution







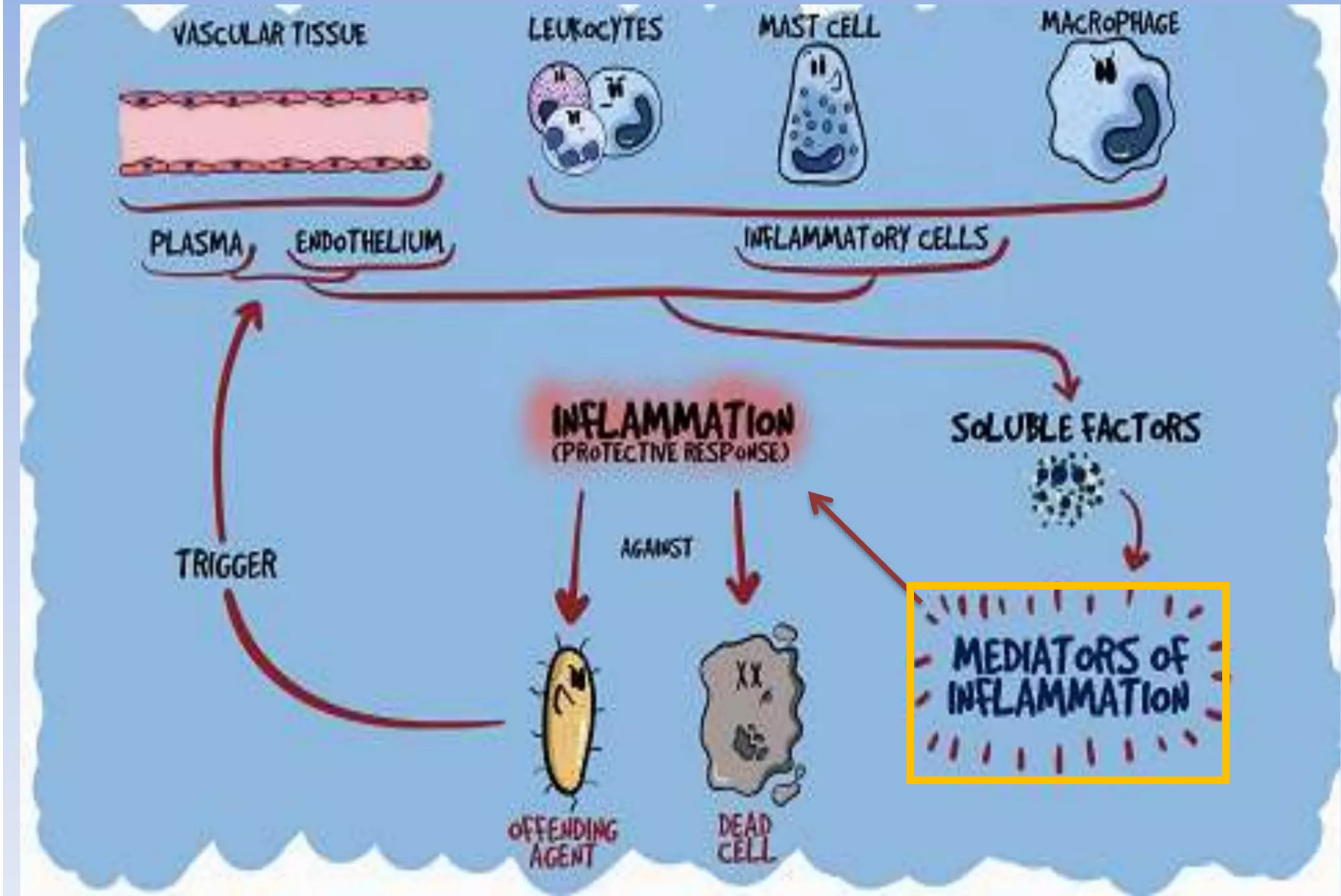

A, NORMAL AXIAL FLOW
B, MARGINATION AND PAVEMENTING
C, ROLLING AND ADHESION
D, EMIGRATION AND DIAPEDESIS

Objectives

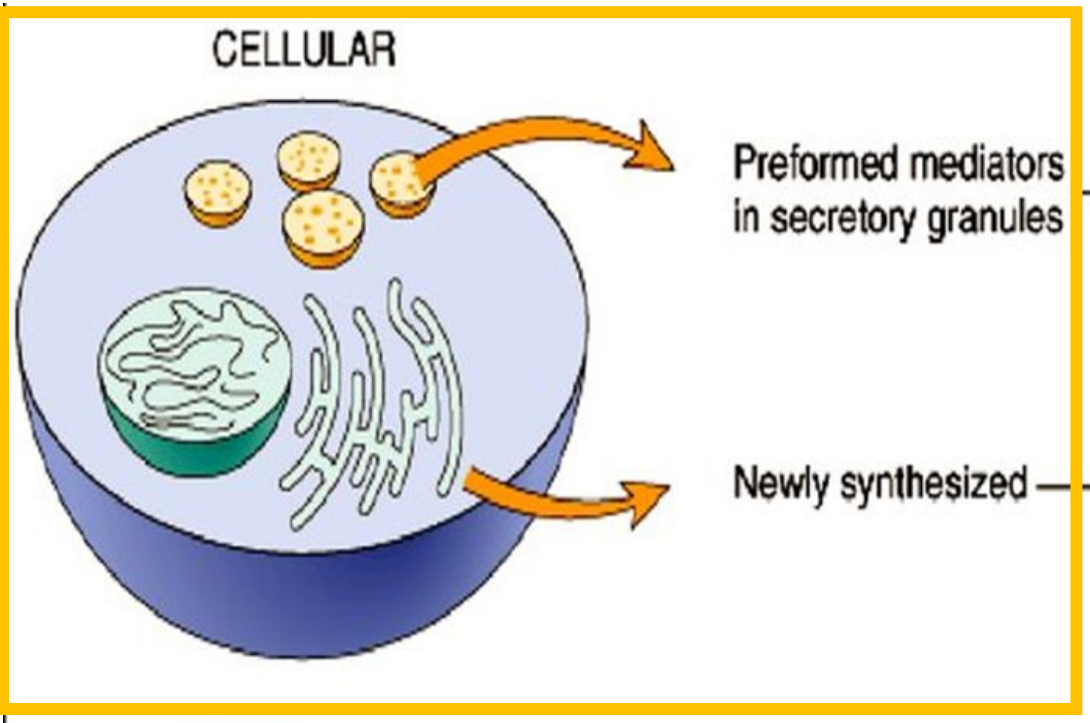


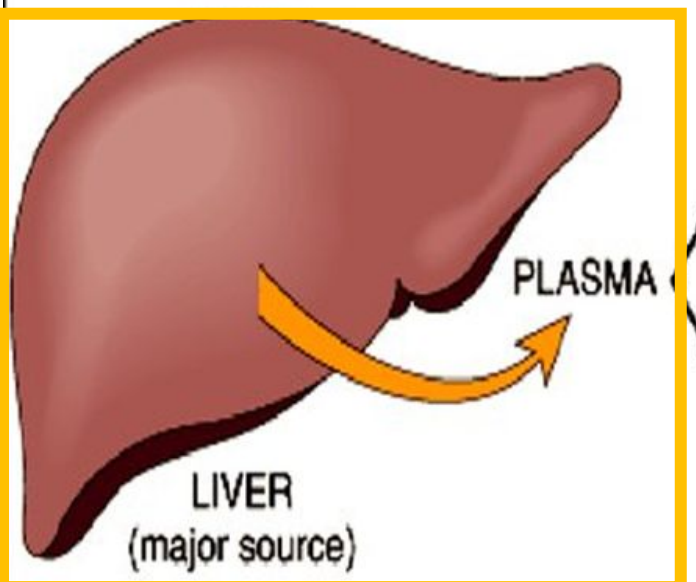
At the end of the lecture first year students should be able to:

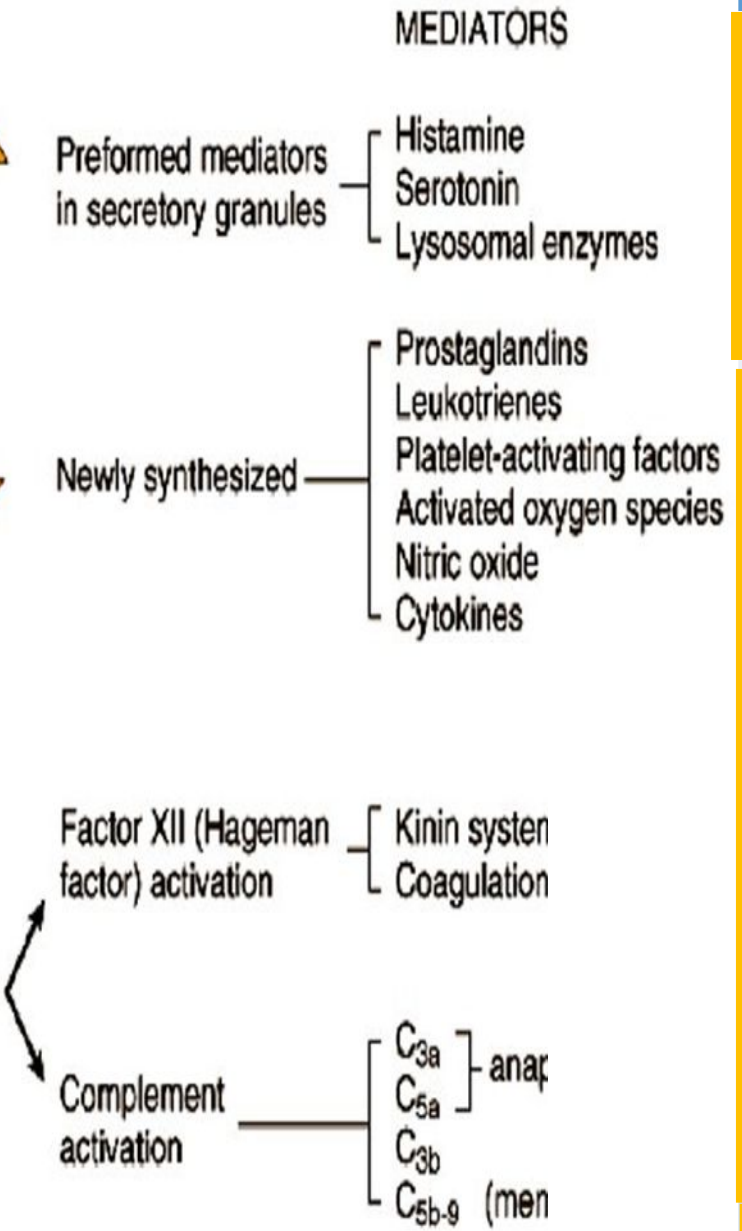
- Name different mediators of inflammation
- Tell the sites of release of different mediators of inflammation
- Understand the functions of different mediators of inflammation
- Enumerate different morphological types of acute inflammation with their examples


VASCULAR TISSUE
LEUKOCYTES
MAST CELL
MACROPHAGE
PLASMA
ENDOTHELIUM
INFLAMMATORY CELLS
INFLAMMATION
(PROTECTIVE RESPONSE)
SOLUBLE FACTORS
TRIGGER
AGAINST
MEDIATORS OF INFLAMMATION
OFFENDING AGENT
DEAD CELL

Mediators of Inflammation







Mediators	Cells
Histamine and Serotonin	Mast cells, Basophils, Platelets
Lysosomal Enzymes and ROS	Macrophages, Neutrophils
Prostaglandins, Platelet activating factor and Leukotrienes	Mast cells, Leucocytes
Nitric Oxide	Macrophages
Cytokines	Macrophages, Endothelial cells, Mast cells
Chemokines	Leucocytes, activated Macrophages
Complement and Kinins	Liver and released in Plasma

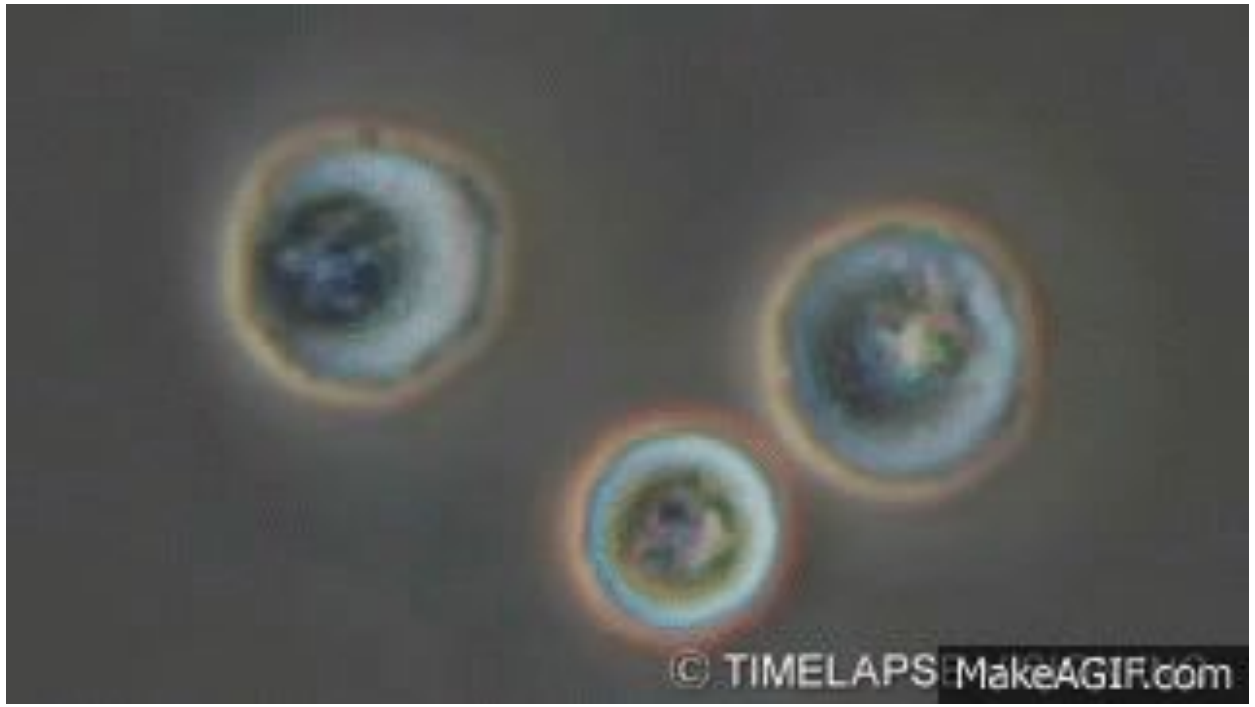
TABLE 2-4 -- The Actions of the Principal Mediators of Inflammation

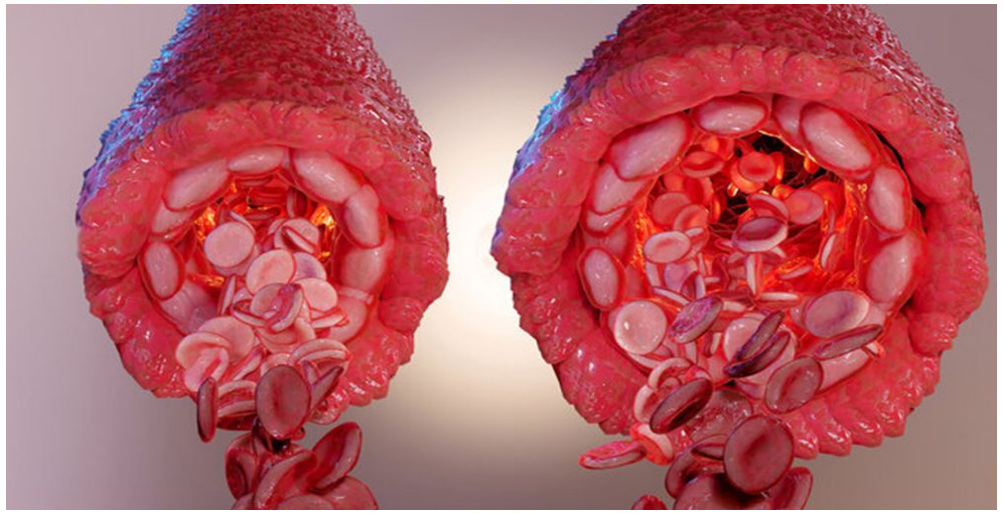
Mediator	Principal Sources	Actions
CELL-DERIVED		
Histamine	Mast cells, basophils, platelets	Vasodilation, increased vascular permeability, endothelial activation
Serotonin	Platelets	Vasodilation, increased vascular permeability
Prostaglandins	Mast cells, leukocytes	Vasodilation, pain, fever
Leukotrienes	Mast cells, leukocytes	Increased vascular permeability, chemotaxis, leukocyte adhesion and activation
Platelet-activating factor	Leukocytes, mast cells	Vasodilation, increased vascular permeability, leukocyte adhesion, chemotaxis, degranulation, oxidative burst
Reactive oxygen species	Leukocytes	Killing of microbes, tissue damage
Nitric oxide	Endothelium, macrophages	Vascular smooth muscle relaxation, killing of microbes
Cytokines (TNF, IL-1)	Macrophages, endothelial cells, mast cells	Local endothelial activation (expression of adhesion molecules), fever/pain/anorexia/hypotension, decreased vascular resistance (shock)
Chemokines	Leukocytes, activated macrophages	Chemotaxis, leukocyte activation
PLASMA PROTEIN–DERIVED		
Complement products (C5a, C3a, C4a)	Plasma (produced in liver)	Leukocyte chemotaxis and activation, vasodilation (mast cell stimulation)
Kinins	Plasma (produced in liver)	Increased vascular permeability, smooth muscle contraction, vasodilation, pain
Proteases activated during coagulation	Plasma (produced in liver)	Endothelial activation, leukocyte recruitment

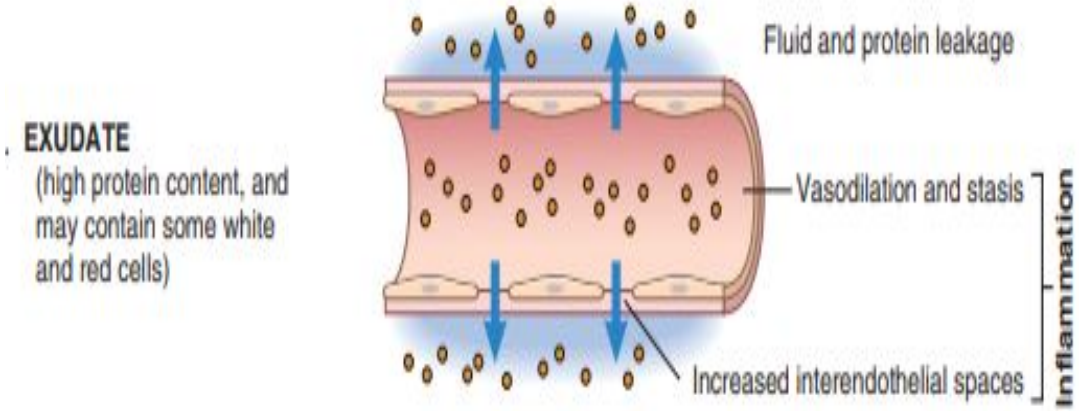
IL-1, interleukin-1; MAC, membrane attack complex; TNF, tumor necrosis factor.

Vasoactive Amides (Histamine and Serotonin)

Histamine	Mast cells, basophils, platelets	Vasodilation, increased vascular permeability, endothelial activation
Serotonin	Platelets	Vasodilation, increased vascular permeability







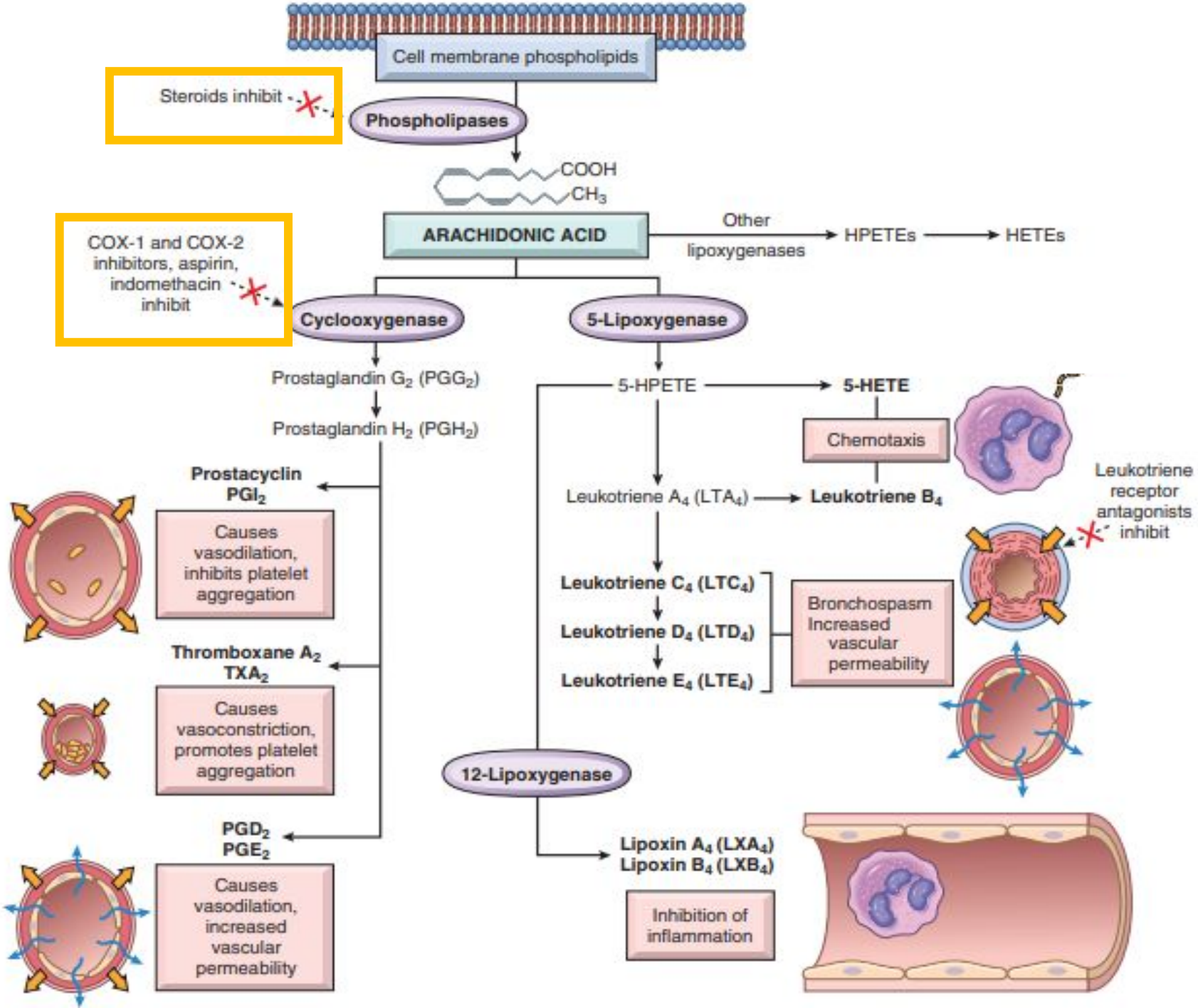
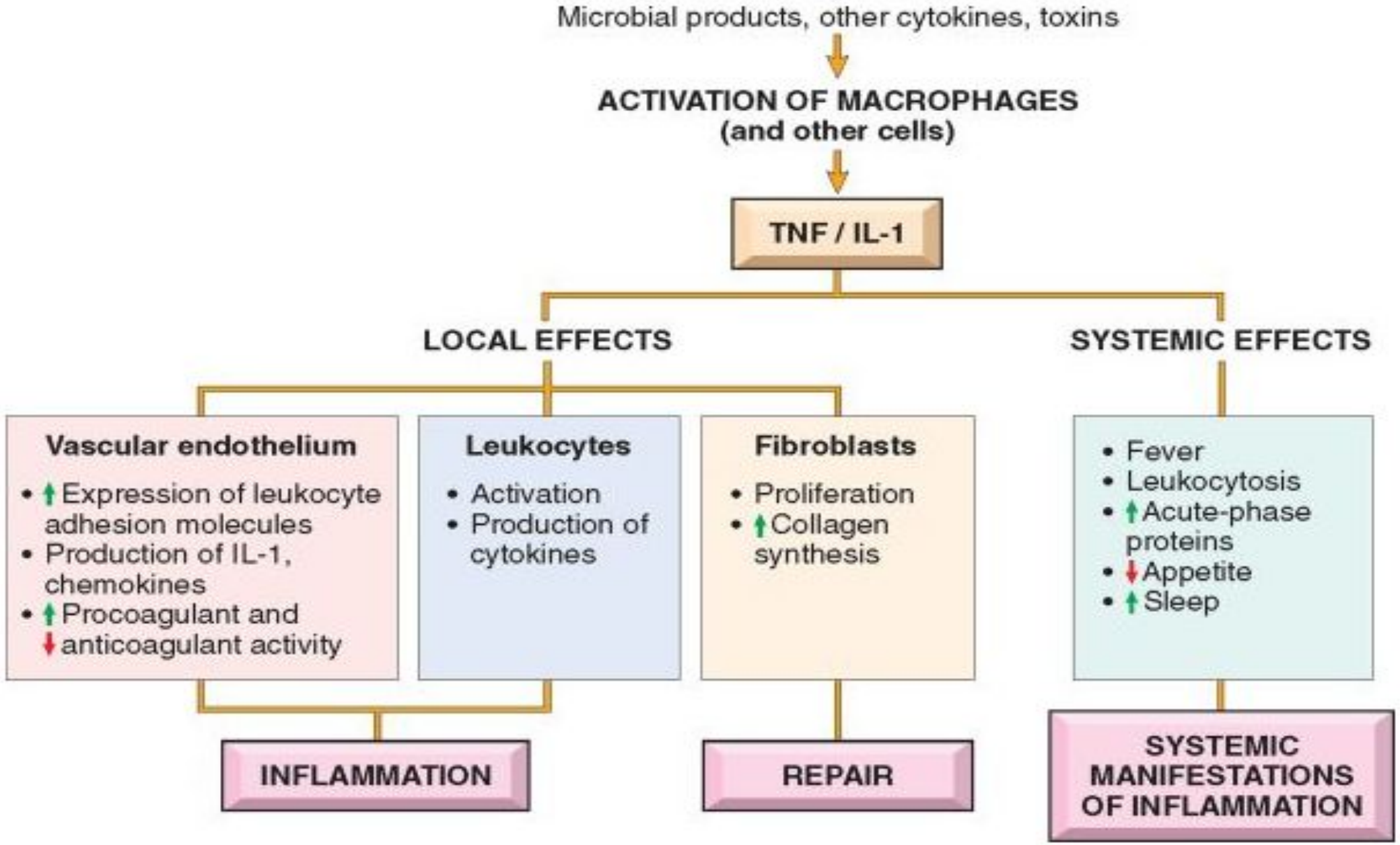

Cell membrane phospholipids
Steroids inhibit
Phospholipases
COOH
CH3
ARACHIDONIC ACID
Other lipoxygenases
HPETEs
HETEs
COX-1 and COX-2 inhibitors, aspirin, indomethacin inhibit
Cyclooxygenase
5-Lipoxygenase
Prostaglandin G2 (PGG2)
Prostaglandin H2 (PGH2)
5-HPETE
5-HETE
Chemotaxis
Leukotriene A4 (LTA4)
Leukotriene B4
Leukotriene receptor antagonists inhibit
Prostacyclin PGI2
Causes vasodilation, inhibits platelet aggregation
Leukotriene C4 (LTC4)
Leukotriene D4 (LTD4)
Leukotriene E4 (LTE4)
Bronchospasm Increased vascular permeability
Thromboxane A2 TXA2
Causes vasoconstriction, promotes platelet aggregation
12-Lipoxygenase
PGD2 PGE2
Causes vasodilation, increased vascular permeability
Lipoxin A4 (LXA4)
Lipoxin B4 (LXB4)
Inhibition of inflammation

TABLE 2-5 -- Principal Inflammatory Actions of Arachidonic Acid Metabolites (Eicosanoids)

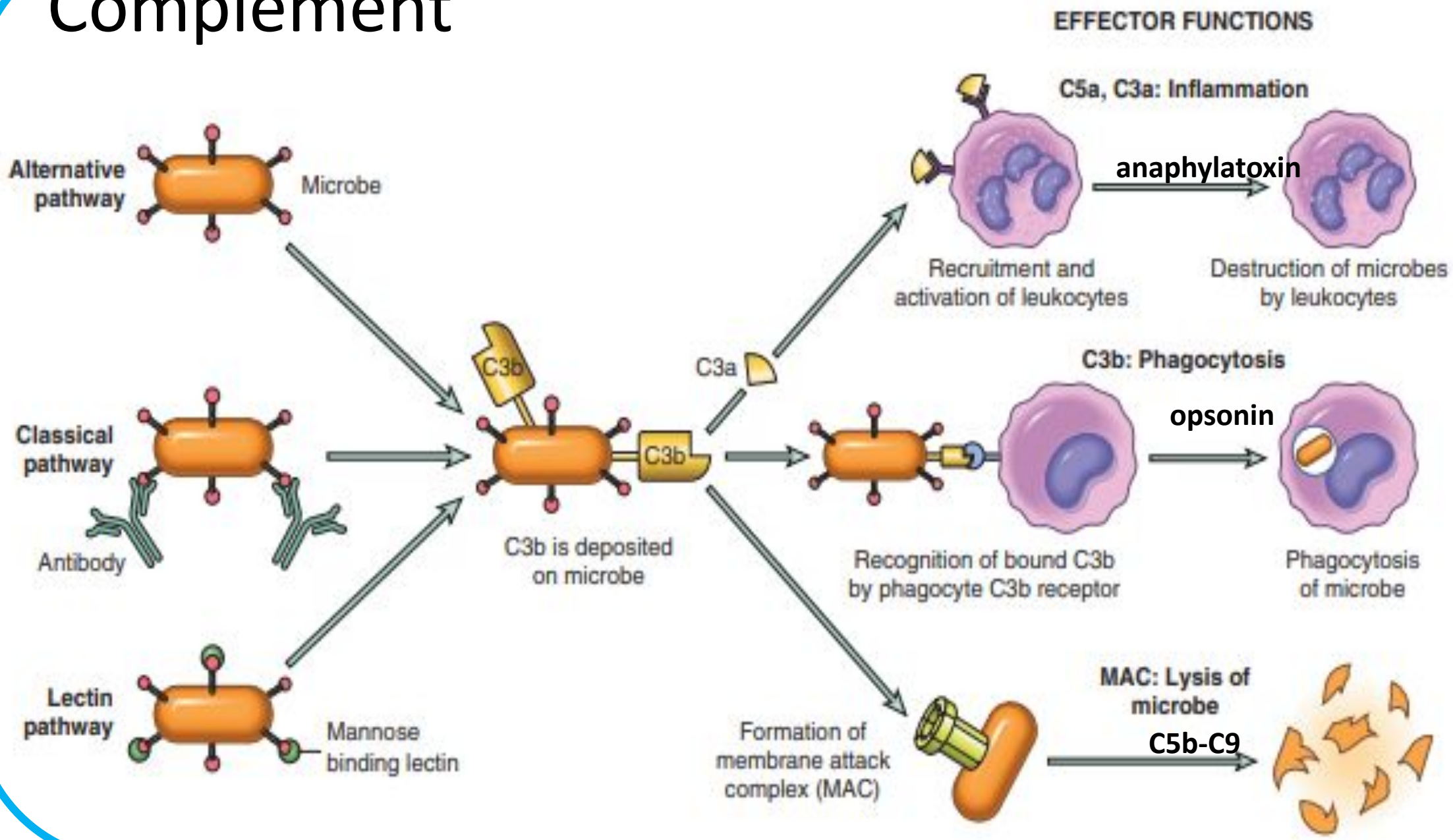
Action	Eicosanoid
Vasodilation	PGI_2 (prostacyclin), PGE_1, PGE_2, PGD_2
Vasoconstriction	Thromboxane A_2, leukotrienes C_4, D_4, E_4
Increased vascular permeability	Leukotrienes C_4, D_4, E_4
Chemotaxis, leukocyte adhesion	Leukotriene B_4, HETE

HETE, hydroxyeicosatetraenoic acid; PGI_2, etc., prostaglandin I_2, etc.

Cytokines (TNF, IL-1)	Macrophages, endothelial cells, mast cells	Local endothelial activation (expression of adhesion molecules), fever/pain, anorexia/hypotension, decreased vascular resistance (shock)



Complement


EFFECTOR FUNCTIONS
C5a, C3a: Inflammation
anaphylatoxin
Recruitment and activation of leukocytes
Destruction of microbes by leukocytes
Alternative pathway
Microbe
C3b
C3a
C3b
C3b: Phagocytosis
opsonin
Classical pathway
Antibody
C3b is deposited on microbe
Recognition of bound C3b by phagocyte C3b receptor
Phagocytosis of microbe
Lectin pathway
Mannose binding lectin
Formation of membrane attack complex (MAC)
MAC: Lysis of microbe
C5b-C9

Function	Mediators
Vasodilatation	Histamine, PAF, Kinins, Prostaglandins
Increased Vascular permeability	Histamine, Leukotrienes, Kinins
Endothelial Activation	Histamine, Cytokines (IL-1,IL-6,TNF)
Pain	Prostaglandins, Kinins, Cytokines (IL-1,IL-6,TNF)
Fever	Prostaglandins, Cytokines (IL-1,IL-6,TNF)
Chemotaxis	Cytokines (IL-1,IL-6,TNF), Chemokines, Complement, PAF, Leukotrienes
Leucocyte activation /adhesion	Leukotrienes, Chemokines, Complement, PAF
Degranulation/Oxidative Burst	PAF
Endothelial activation/Shock	Cytokines (IL-1,IL-6,TNF)
Membrane attack complex	Complement
Smooth muscle contraction	Kinins

Morphological Patterns of Acute inflammation

Morphological Patterns of Acute inflammation

Serous Inflammation

Fibrinous Inflammation

Purulent (Suppurative)

Ulcers

Pseudomembranous Inflammation

Serous Inflammation

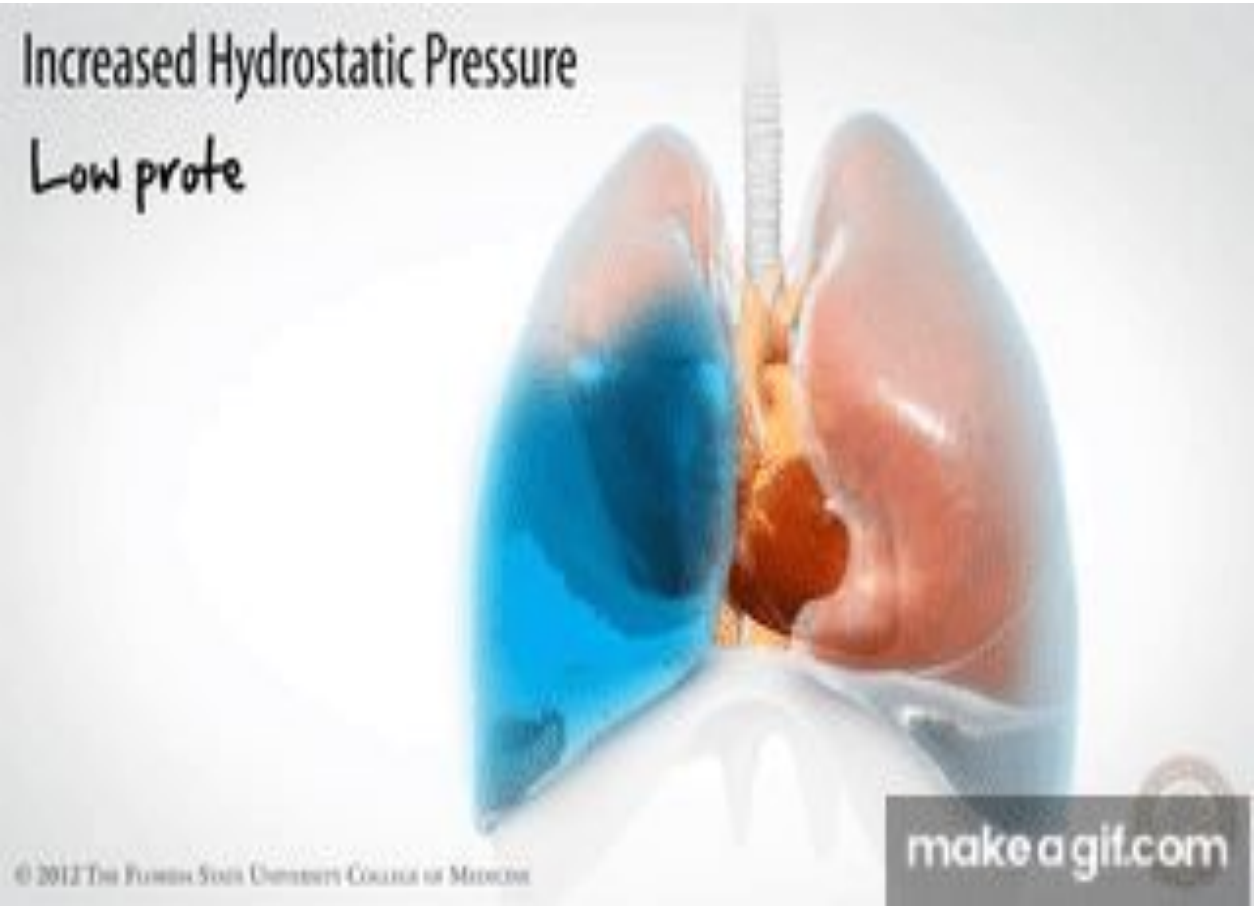
- Serous inflammation is marked by the exudation of cell poor fluid (Transudate) into spaces
- When occur in body cavities termed as EFFUSSION

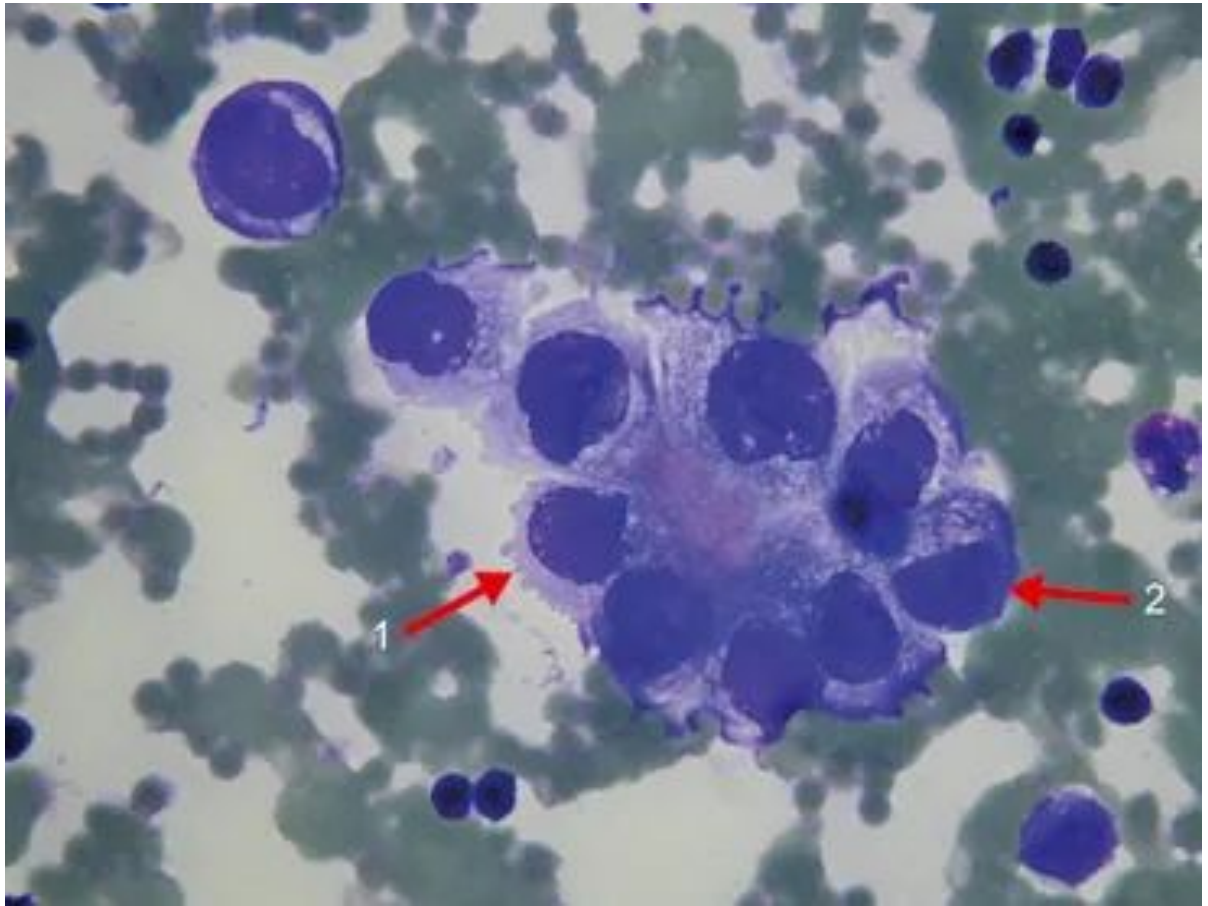




Serous Inflammation

- Fluid may be derived from the plasma (as a result of increased vascular permeability) or
- Secretions of mesothelial cells (as a result of local irritation)





Serous Inflammation

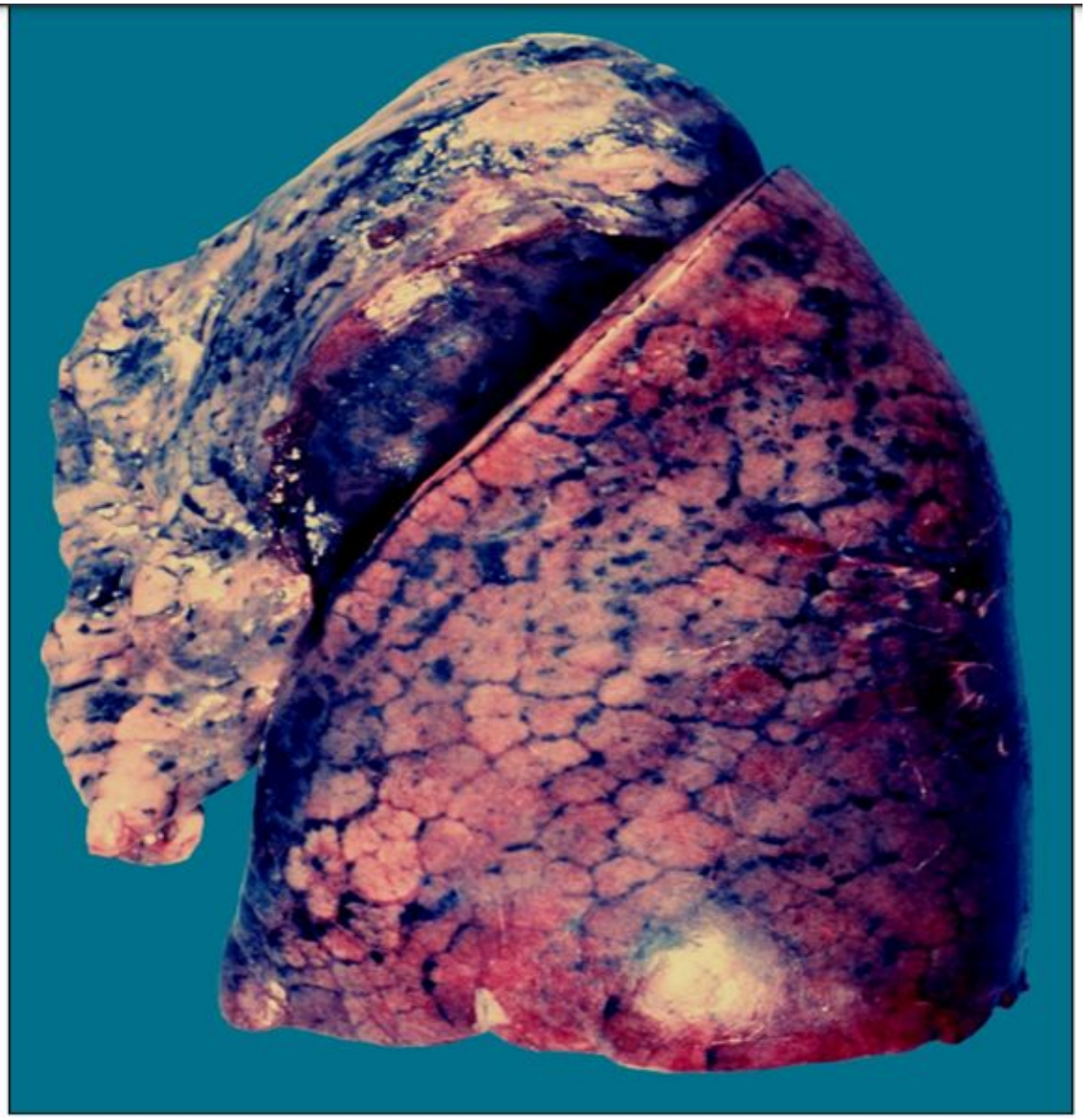
- The skin blister resulting from a burn or viral infection represents accumulation of serous fluid within or immediately beneath the damaged epidermis of the skin

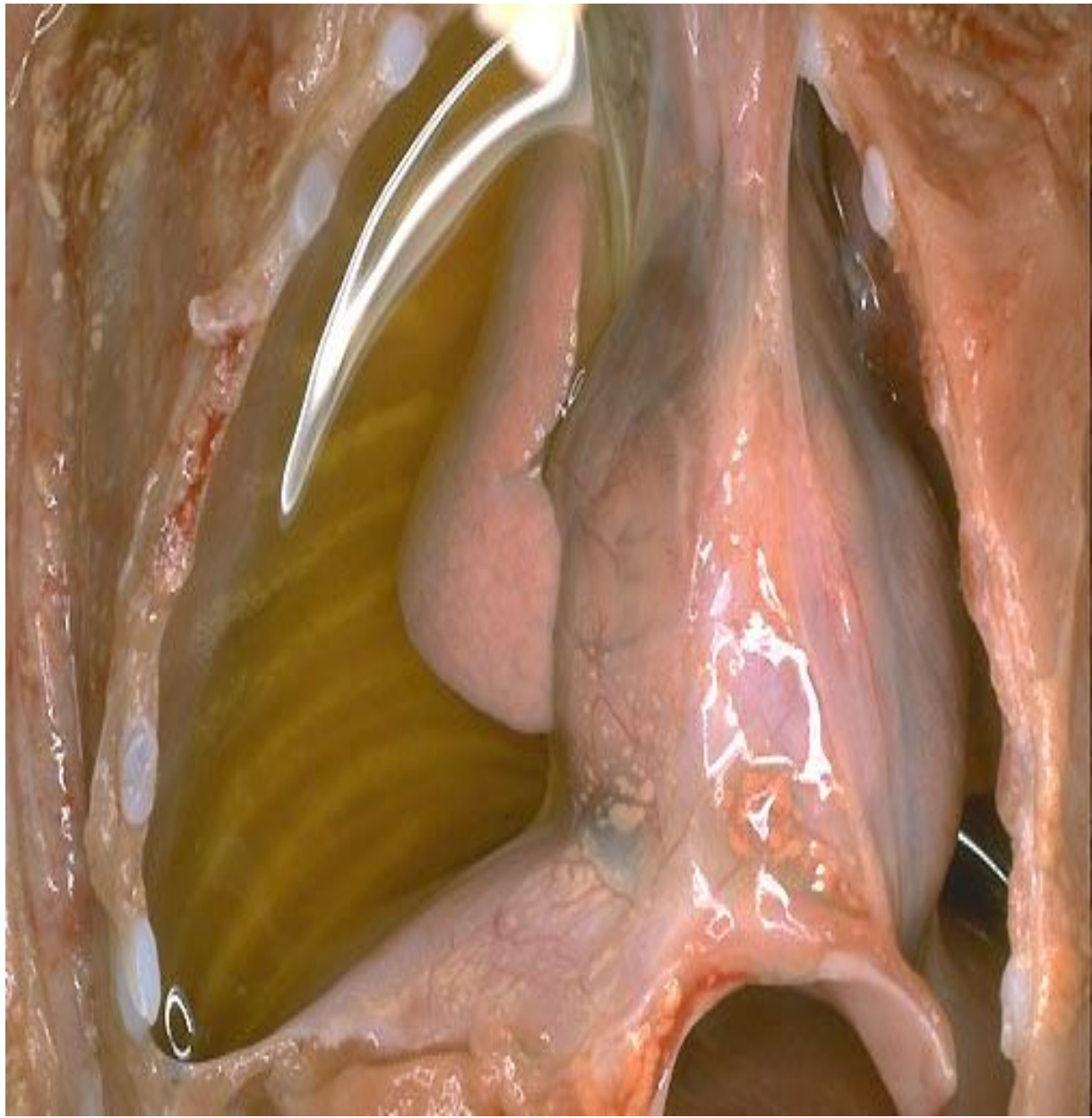




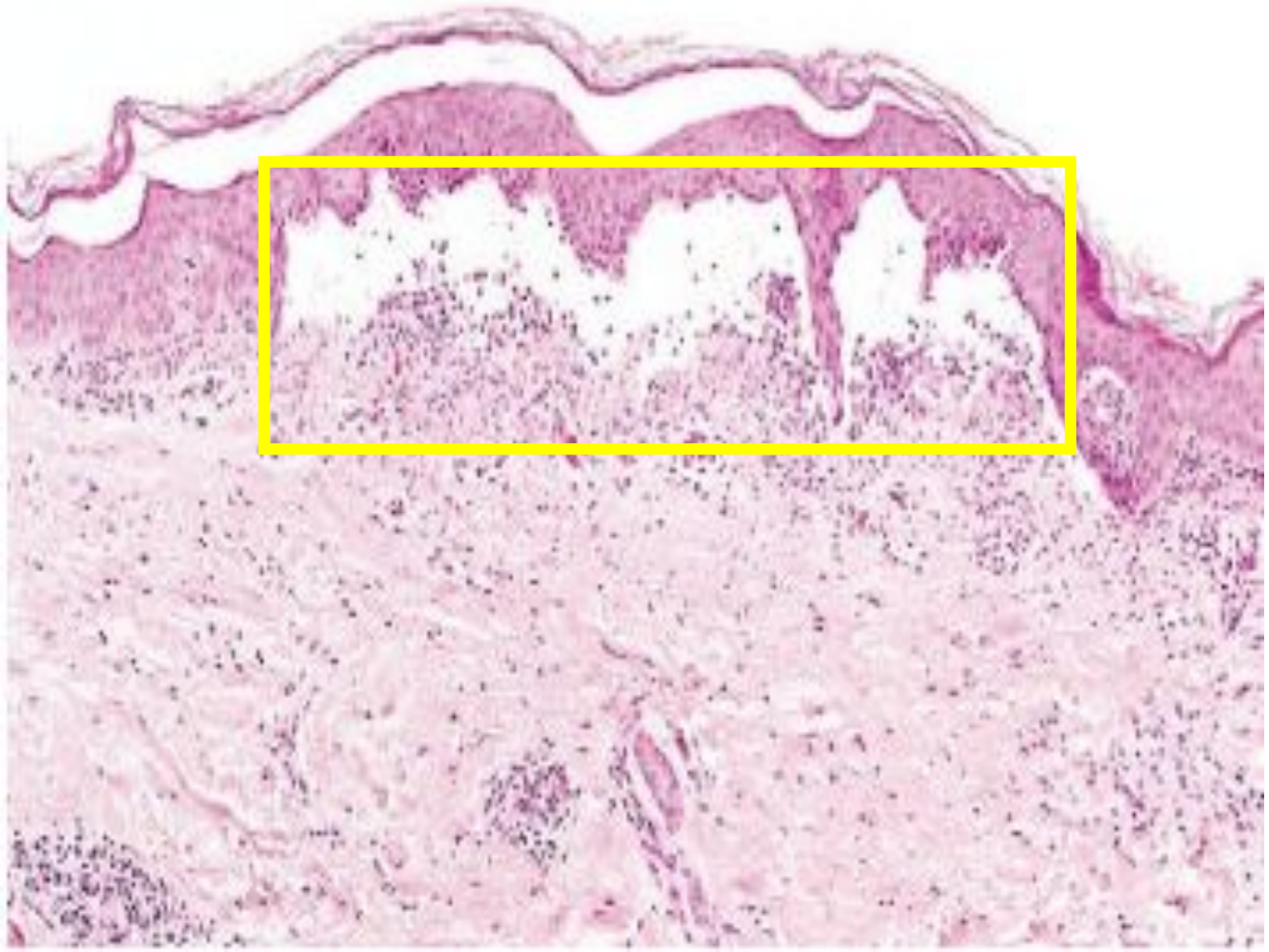


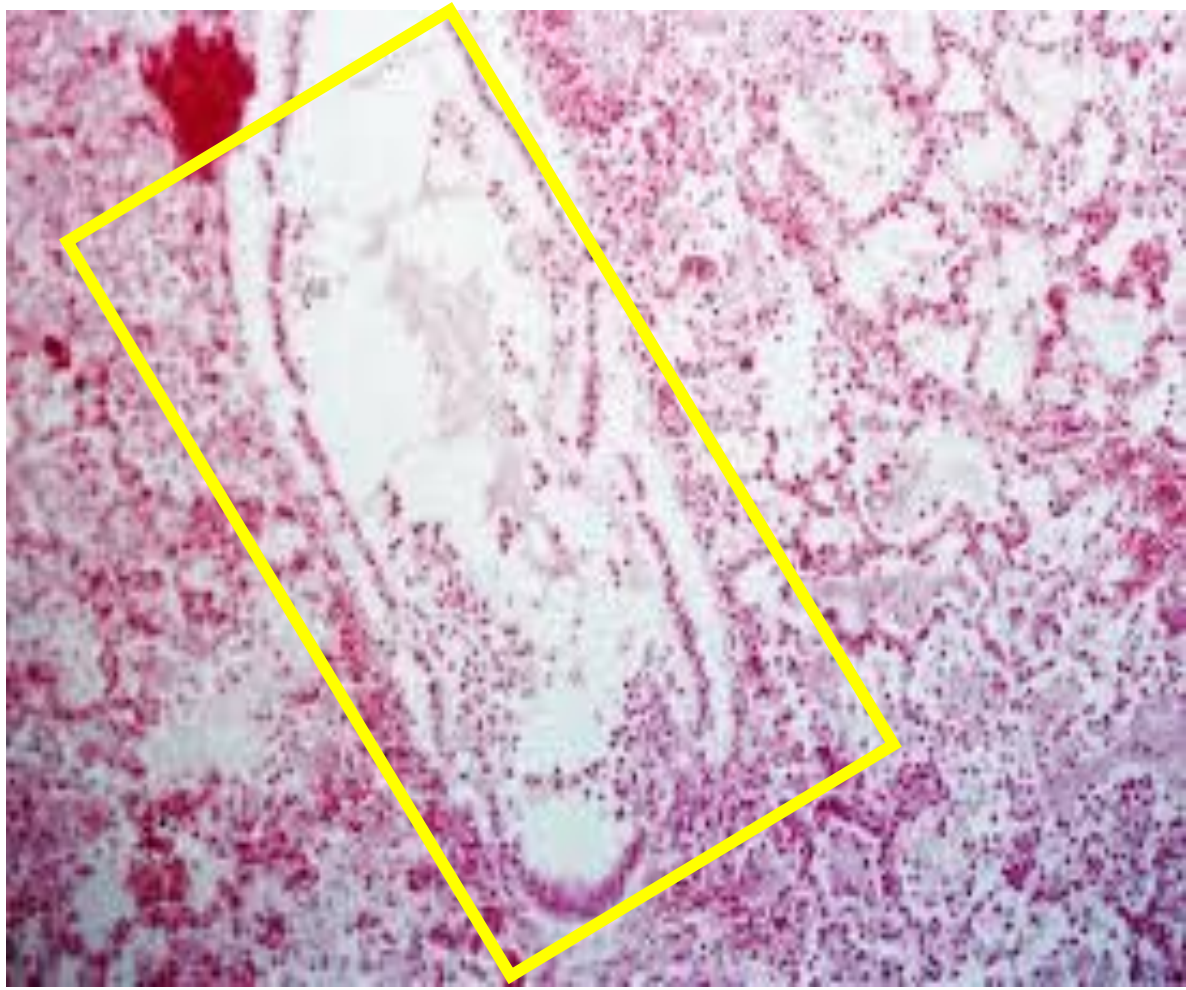
Serous Inflammation





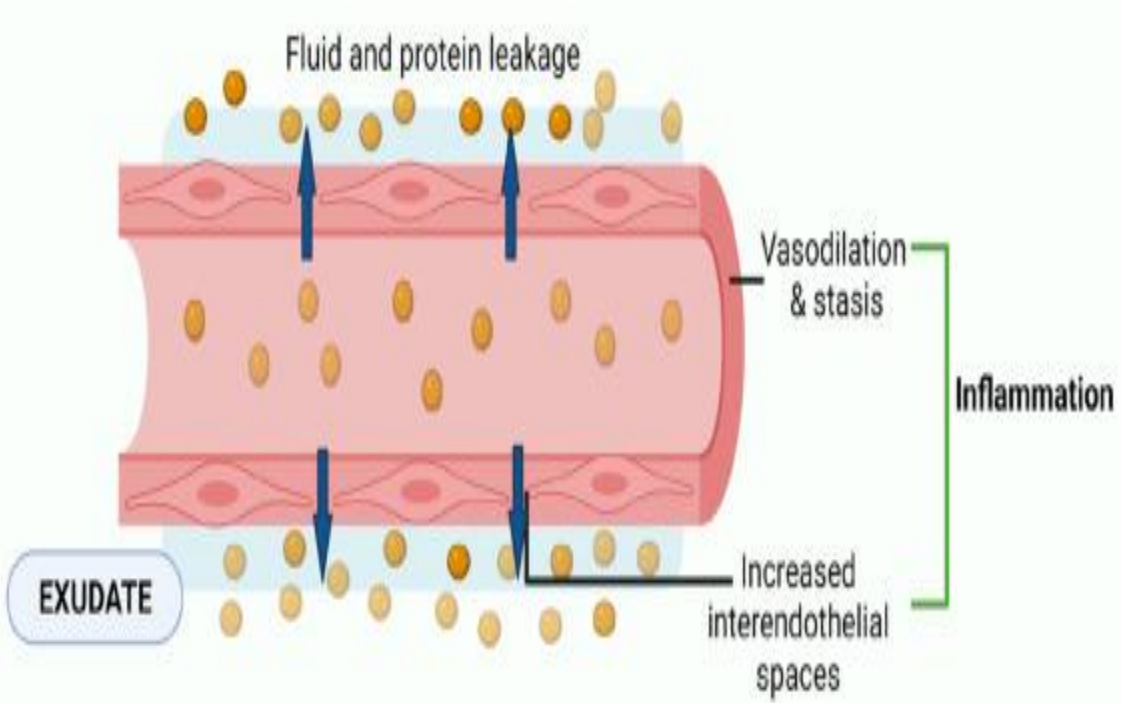
Serous Inflammation

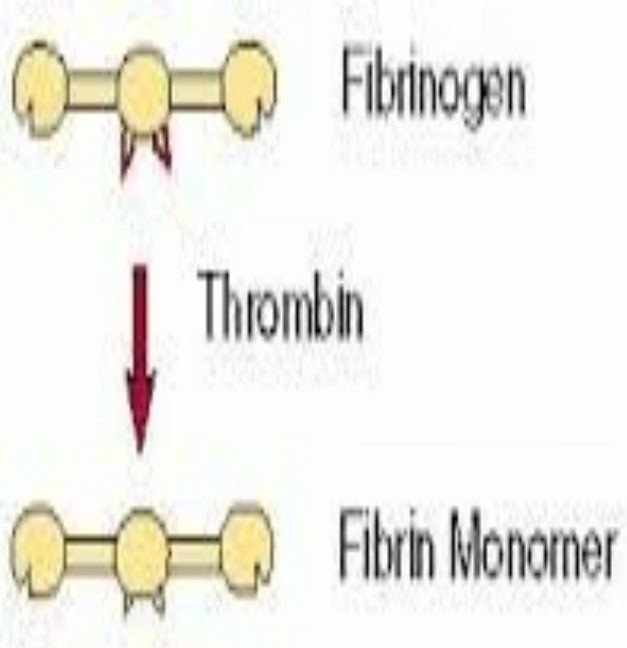


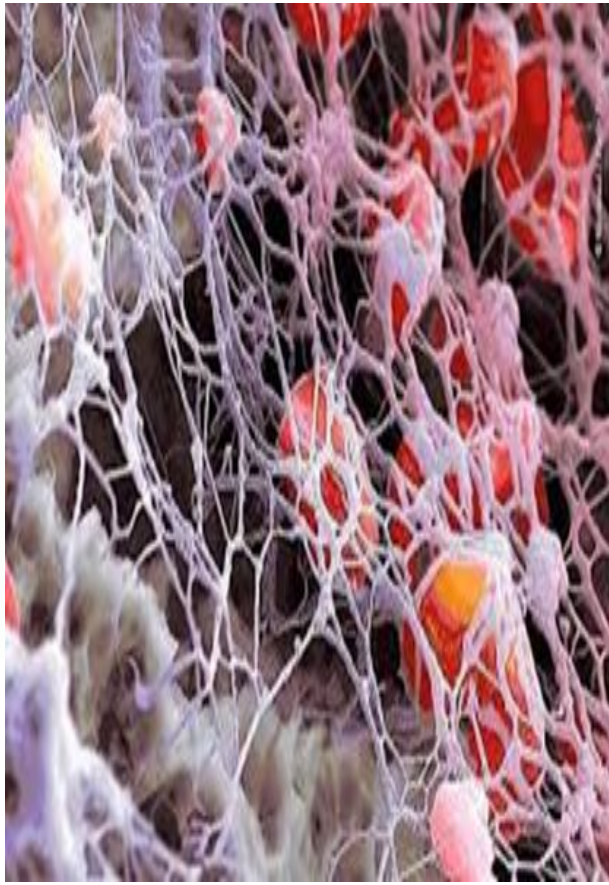


Fibrinous Inflammation

- With greater increase in vascular permeability, large molecules such as fibrinogen pass out of the blood
- This fibrinogen is converted into fibrin which is deposited in the extracellular space

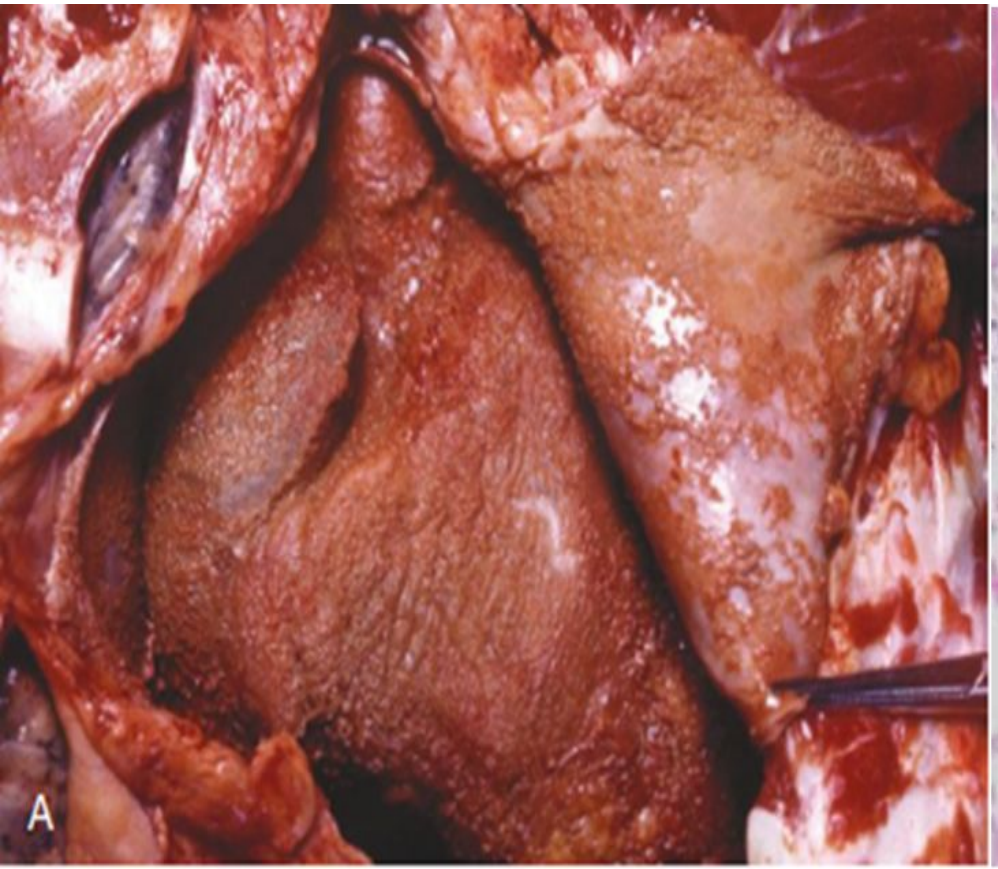


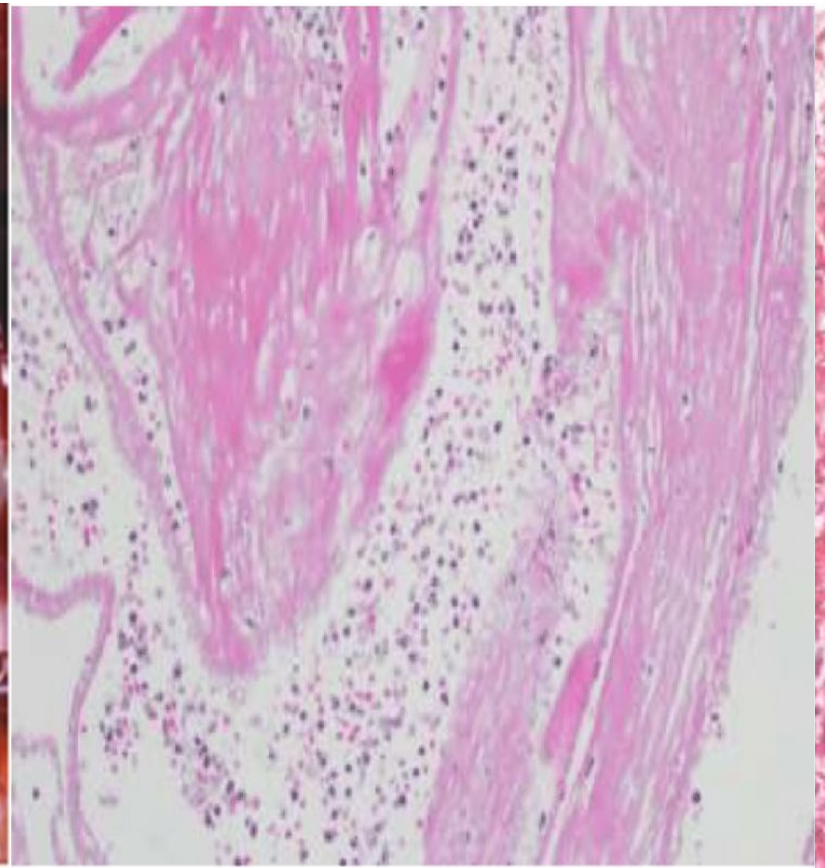


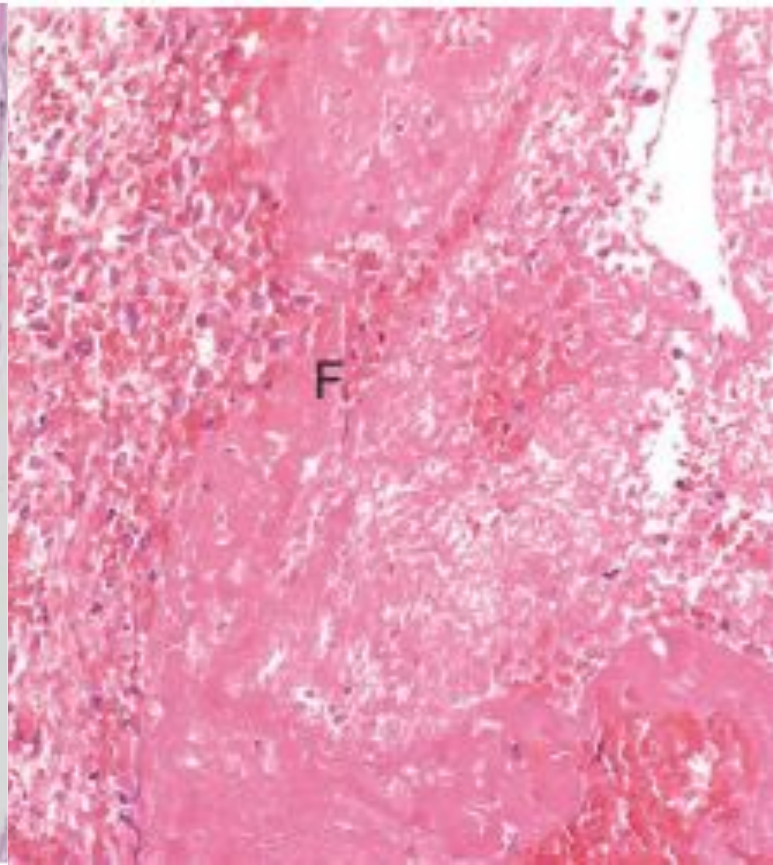


Fibrinous Inflammation

- A fibrinous exudate is characteristic of inflammation in the lining of body cavities, such as the meninges, pericardium and pleura
- Histologically, fibrin appears as an eosinophilic meshwork of threads or sometimes as an amorphous coagulum

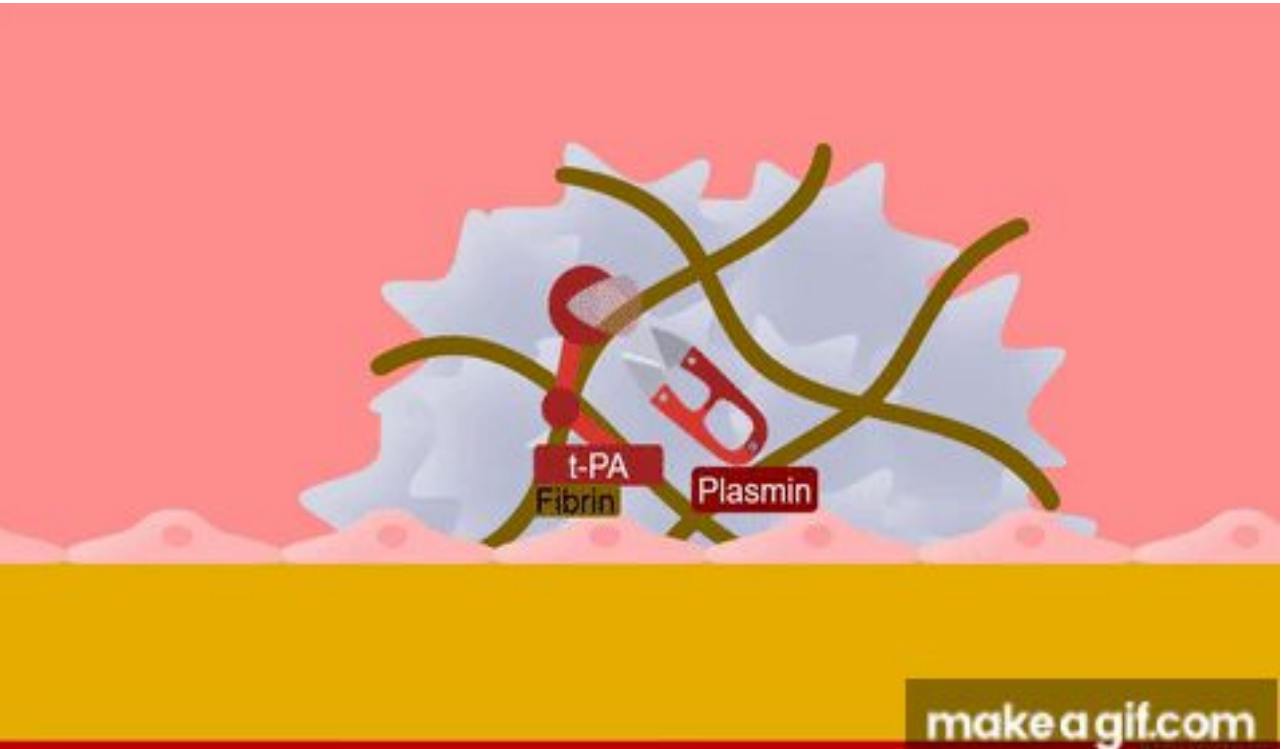






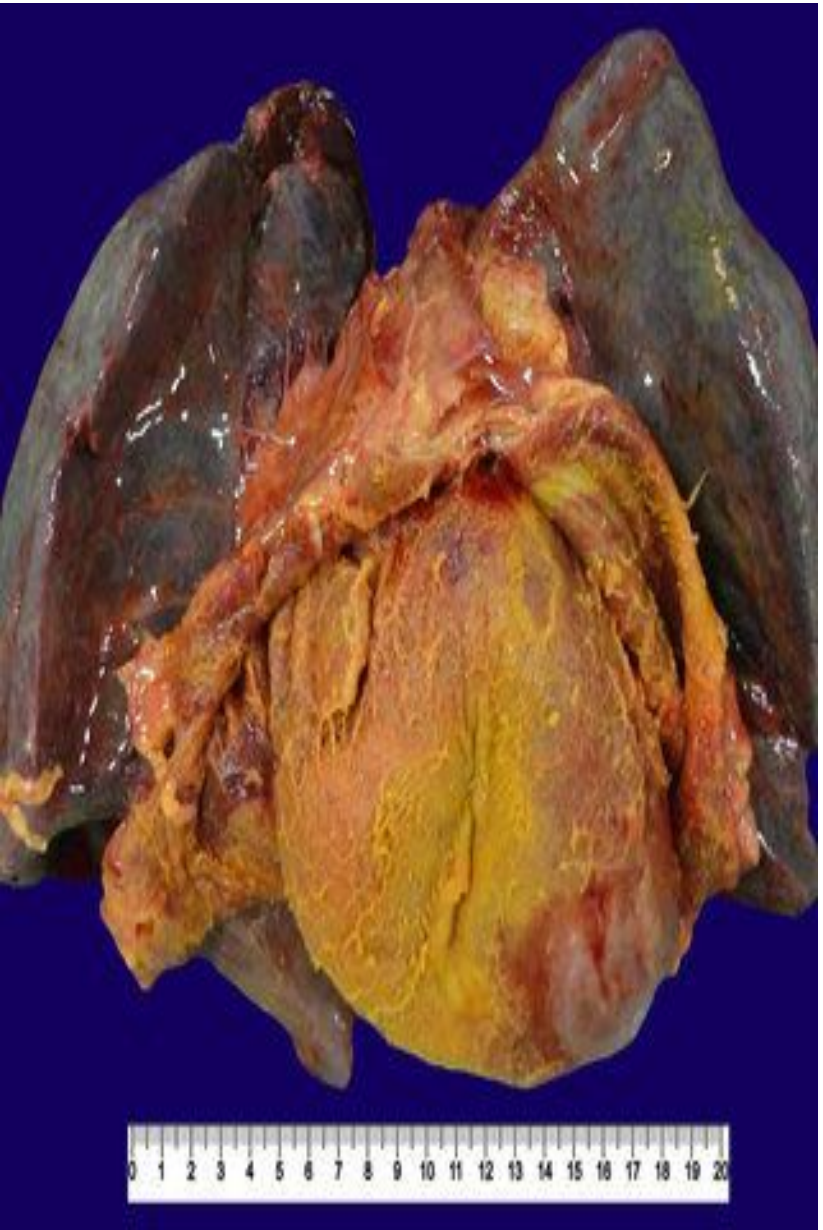
Fibrinous Inflammation

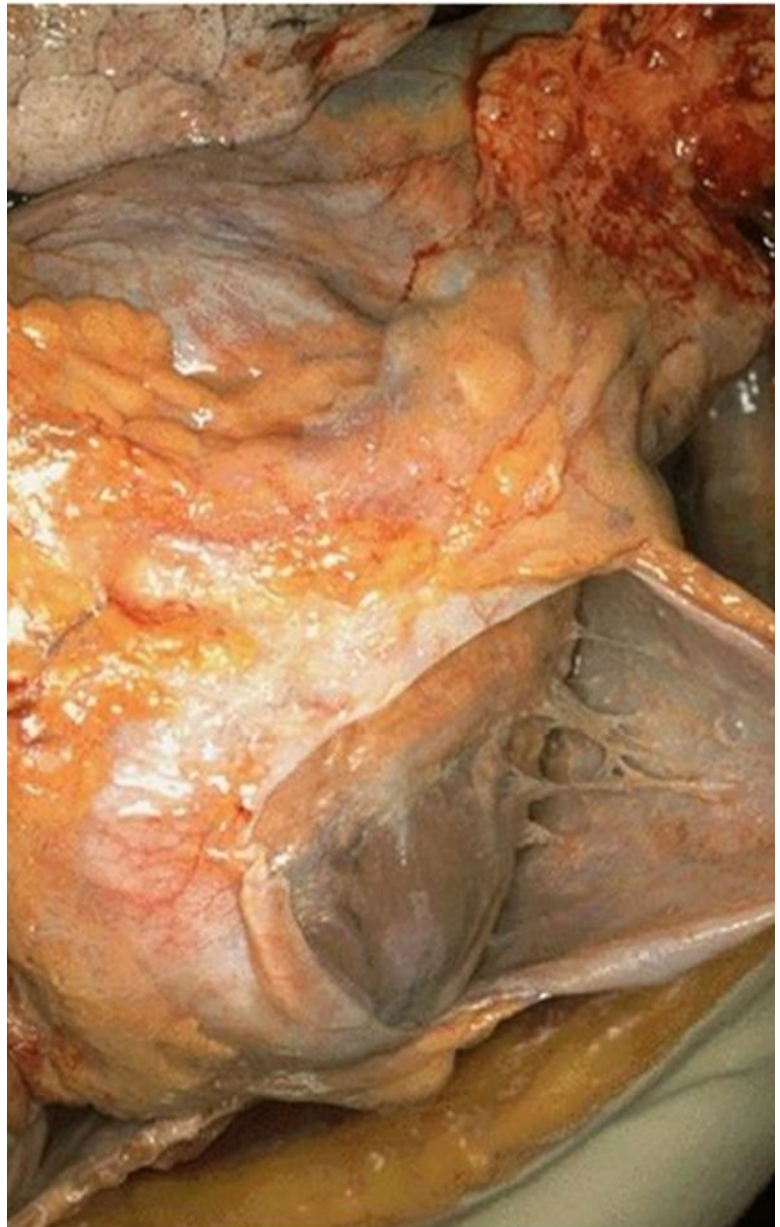
- Fibrinous exudates may be dissolved by fibrinolysis and cleared by macrophages
- If the fibrin is not removed, over time it may stimulate the growth of fibroblasts and blood vessels and thus lead to scarring (organization)





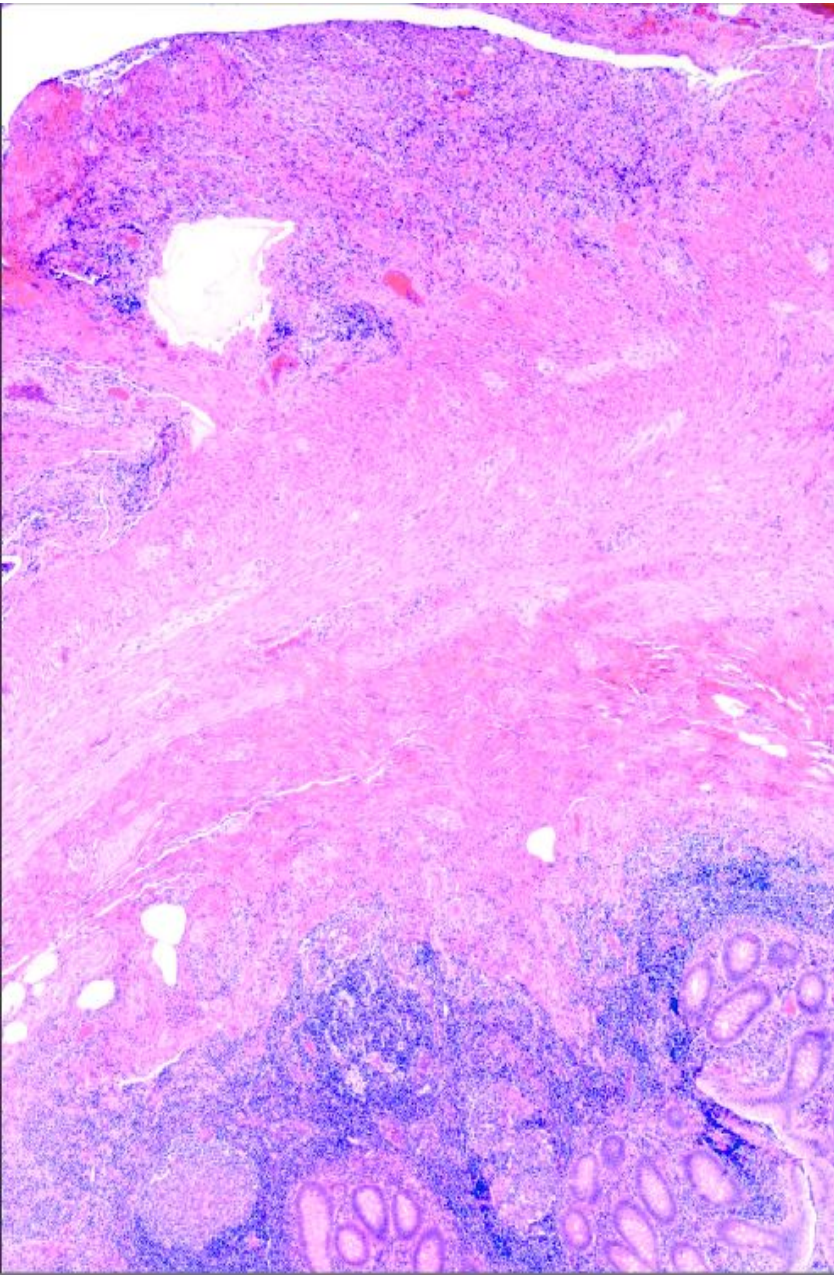
Fibrinous Inflammation

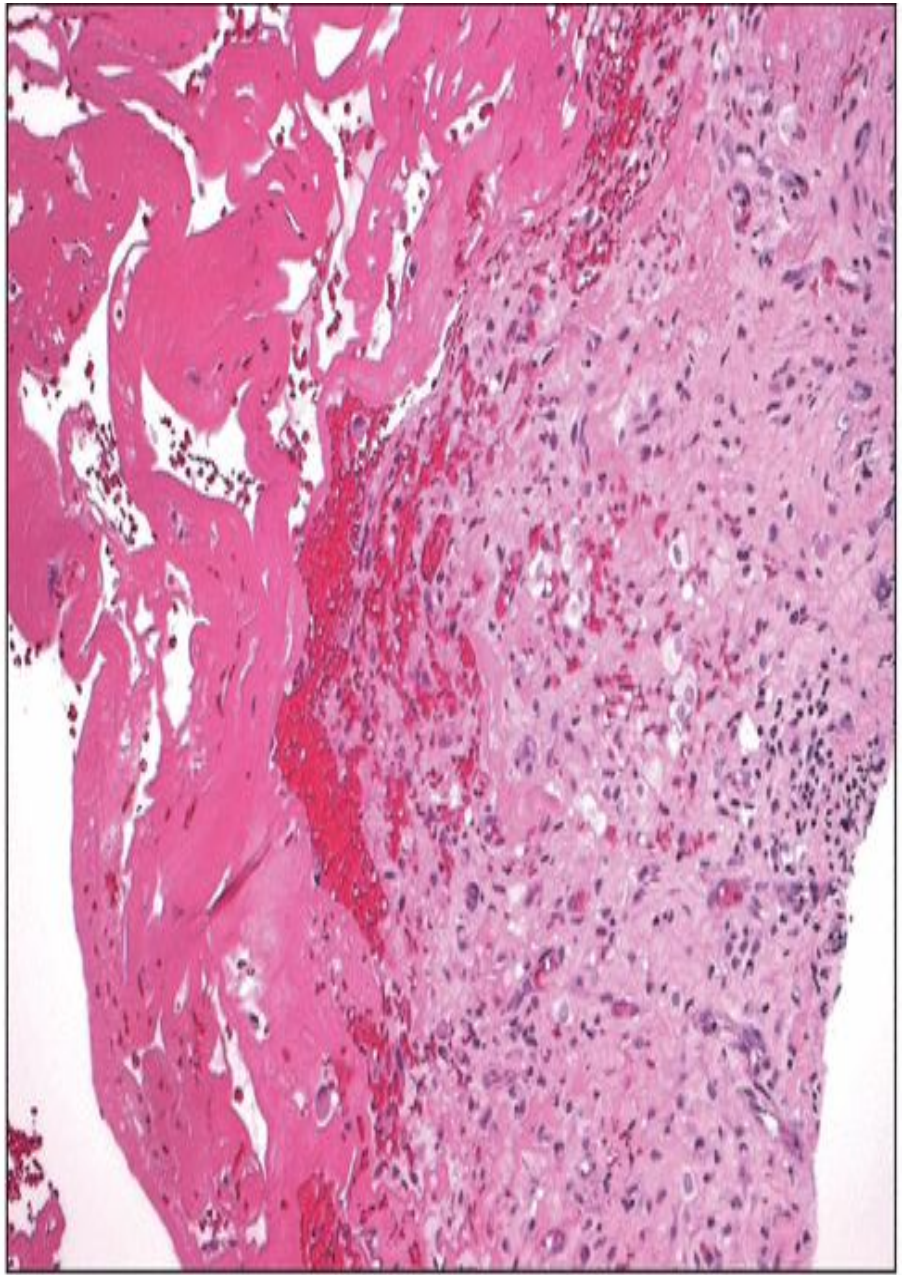


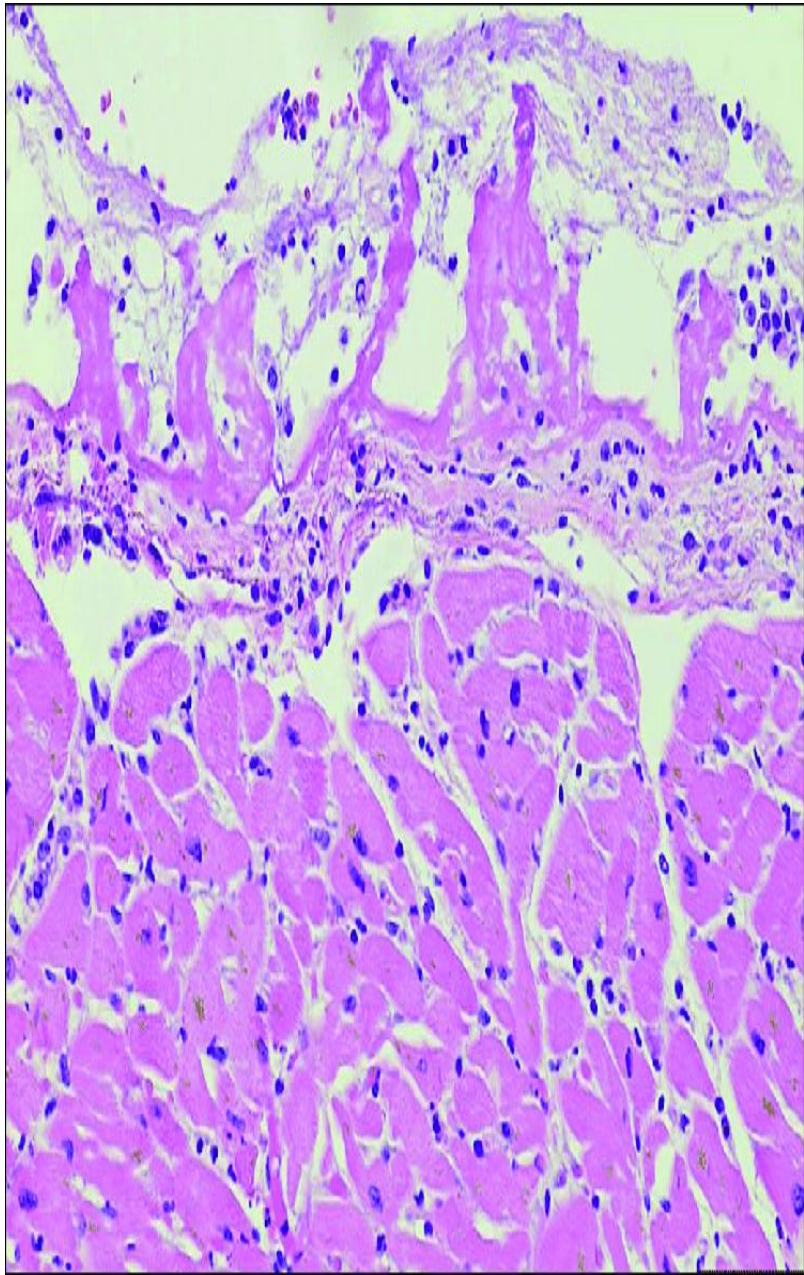




Fibrinous Inflammation

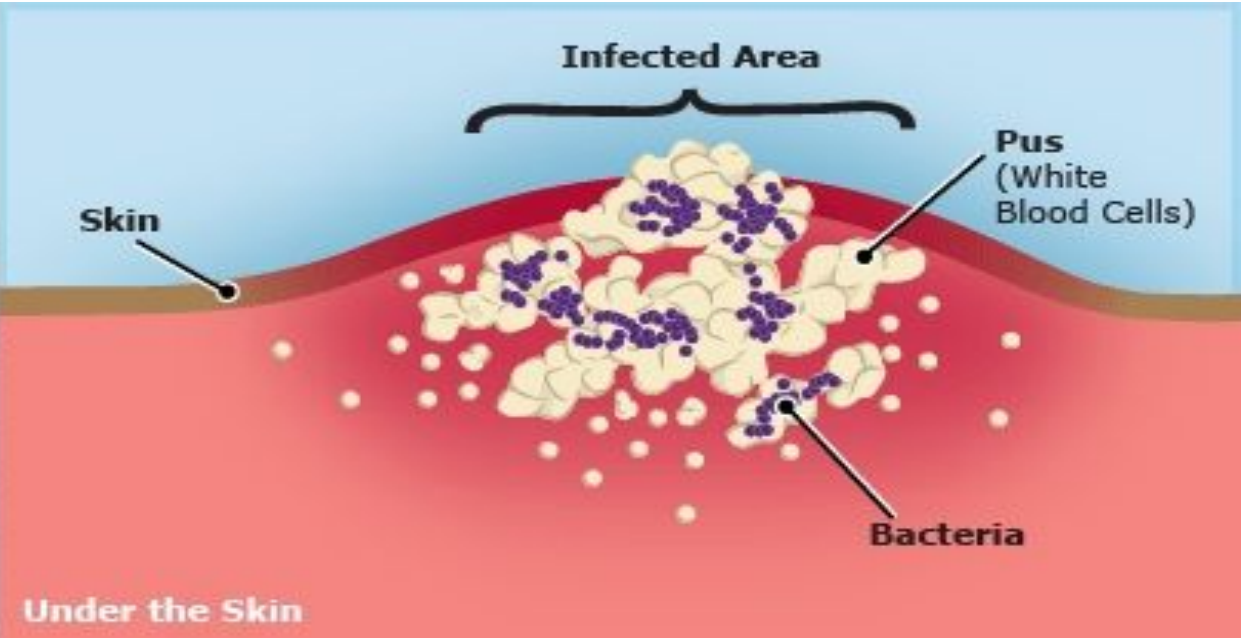






Purulent (Suppurative) Inflammation/ Abscess

- Purulent inflammation is characterized by the production of pus, an exudate consisting of neutrophils, the liquefied debris of necrotic cells, and edema fluid.
- Abscesses are localized collections of purulent inflammatory tissue caused by suppuration buried in a tissue, an organ, or a confined space produced by seeding of pyogenic bacteria into a tissue



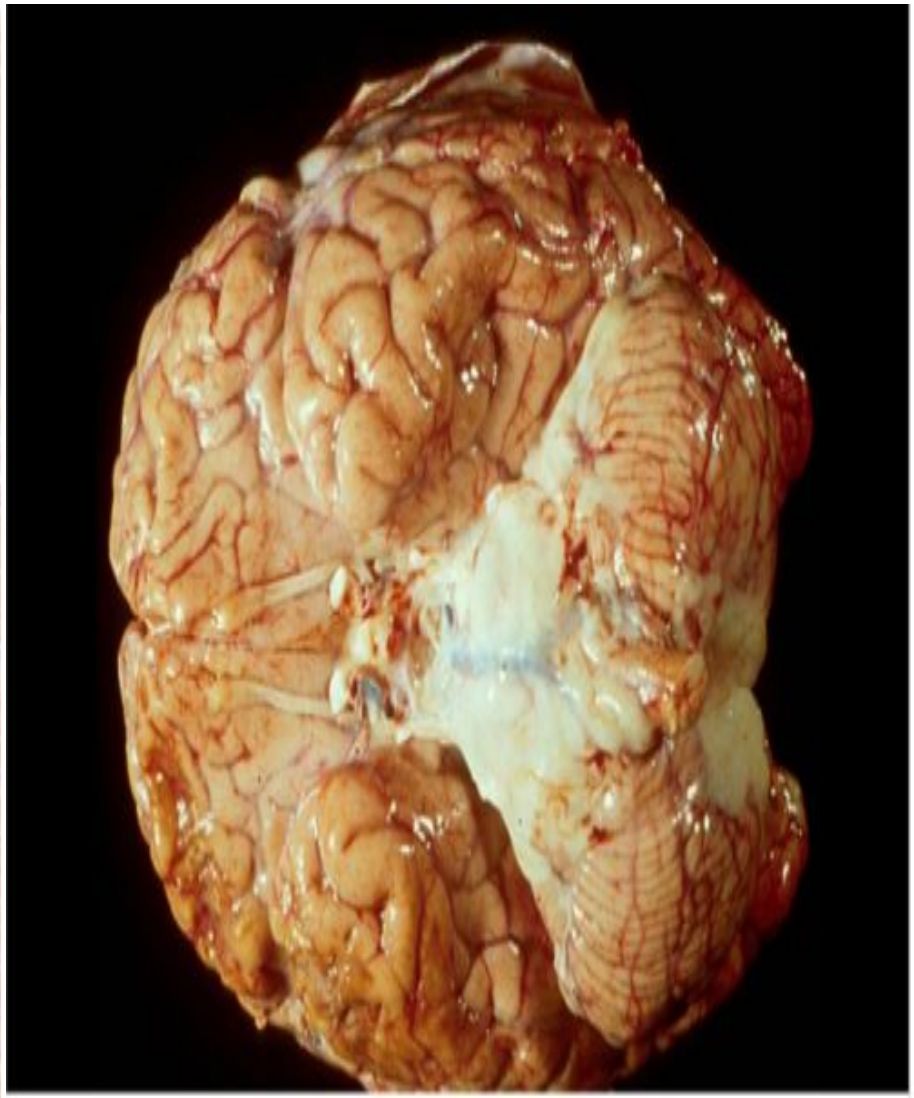




Purulent (Suppurative) Inflammation/Abscess

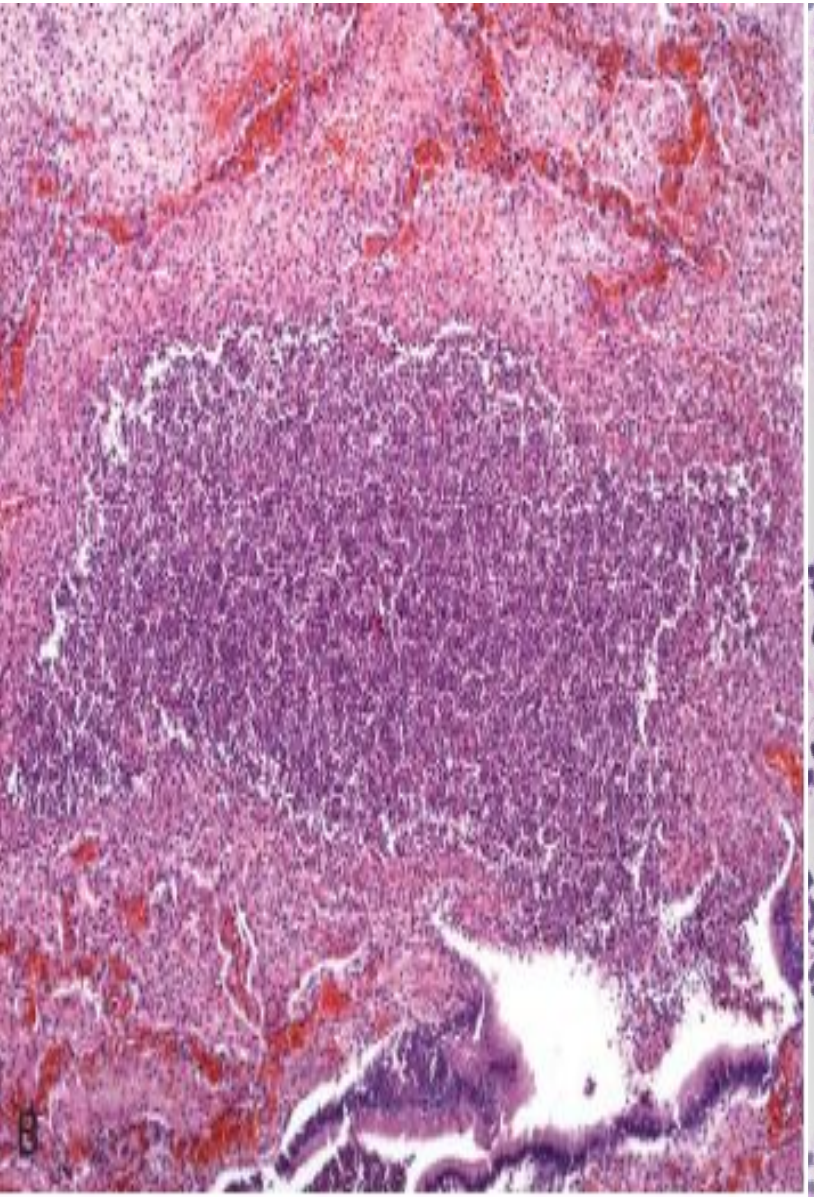


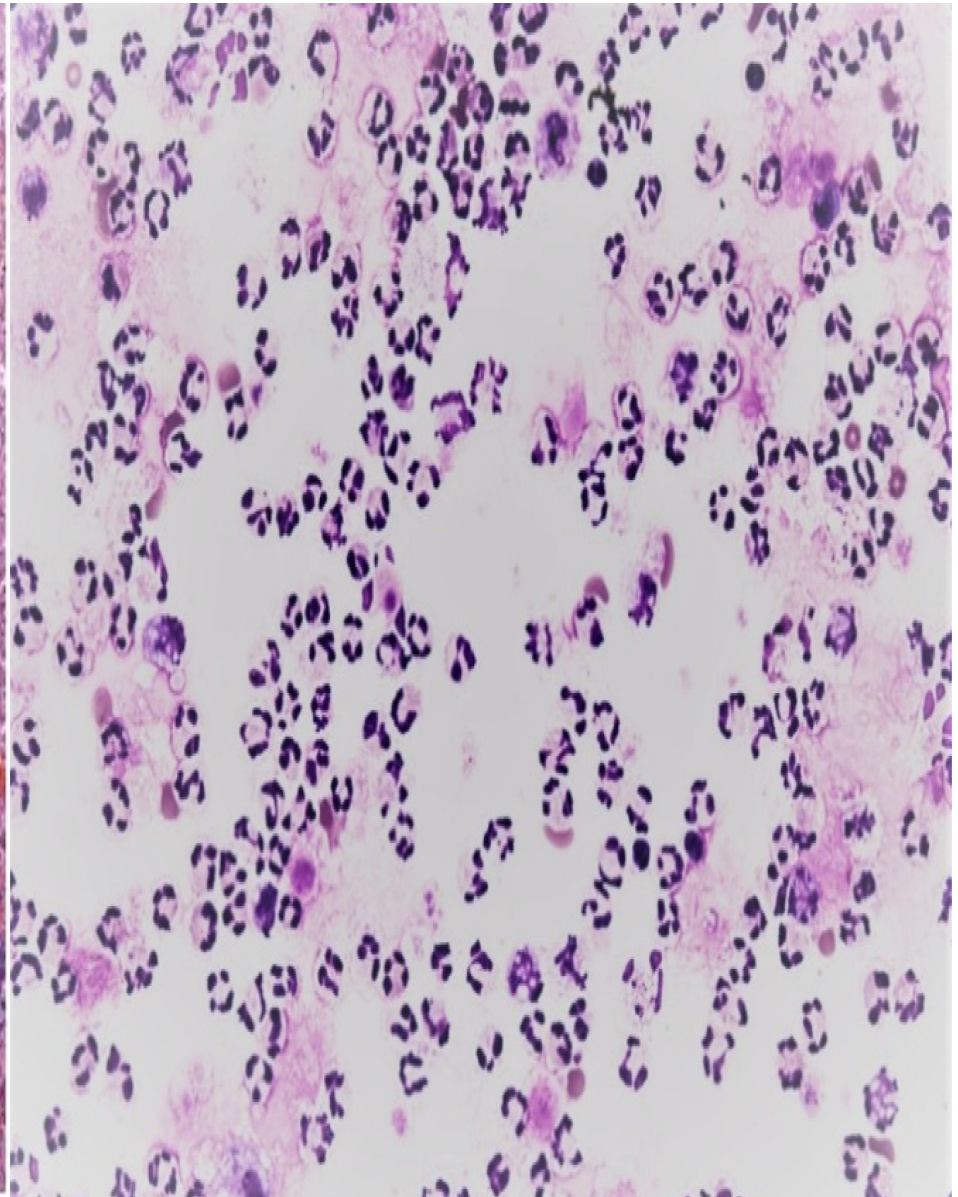


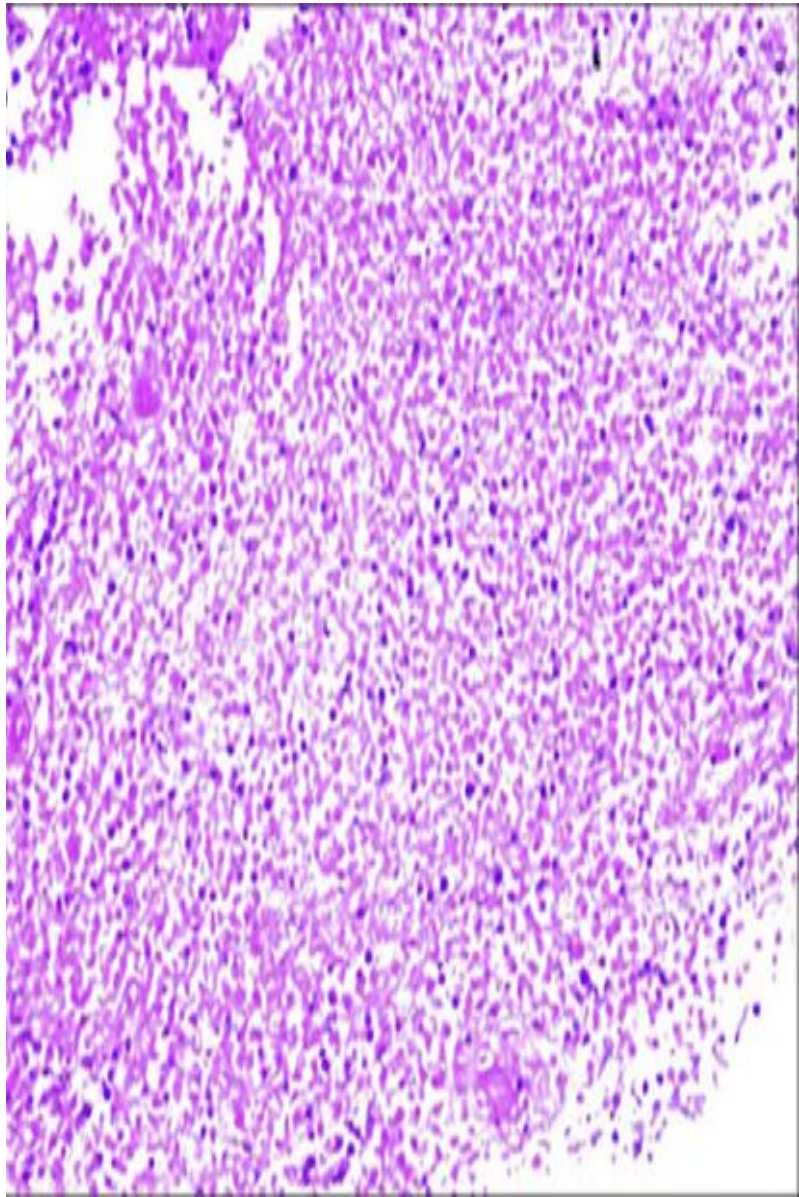


Gross: Acute, purulent meningitis. Note the exudate is in the subarachnoid space. Basal (base of brain) location suggests certain etiologic organisms, e.g. H. Influenza, tuberculosis.

Purulent (Suppurative) Inflammation, Abscess

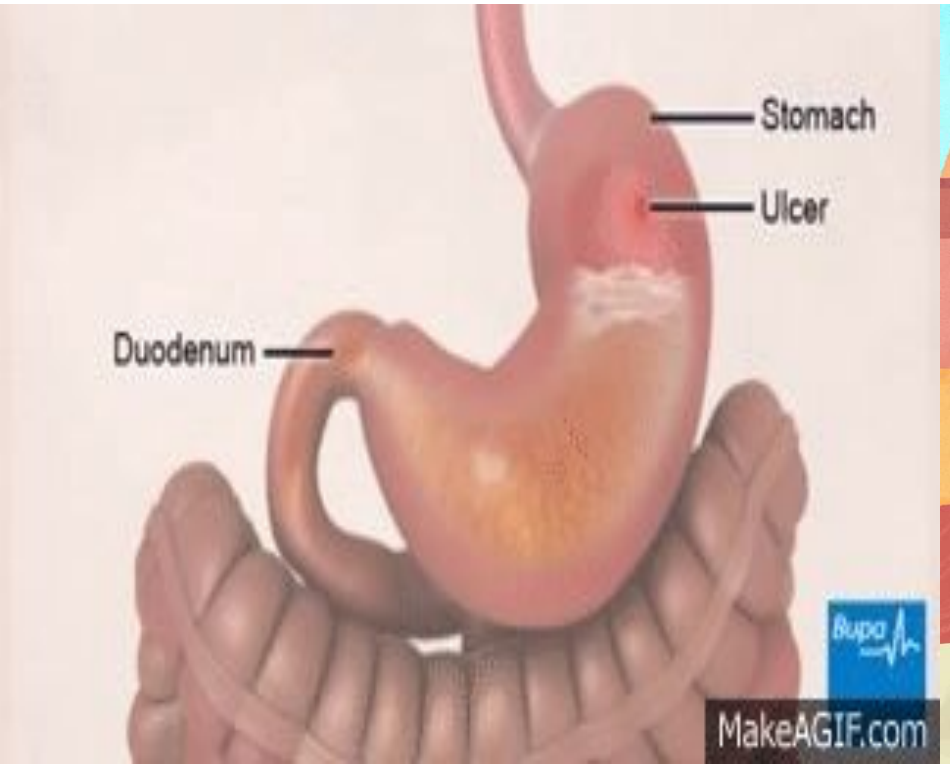


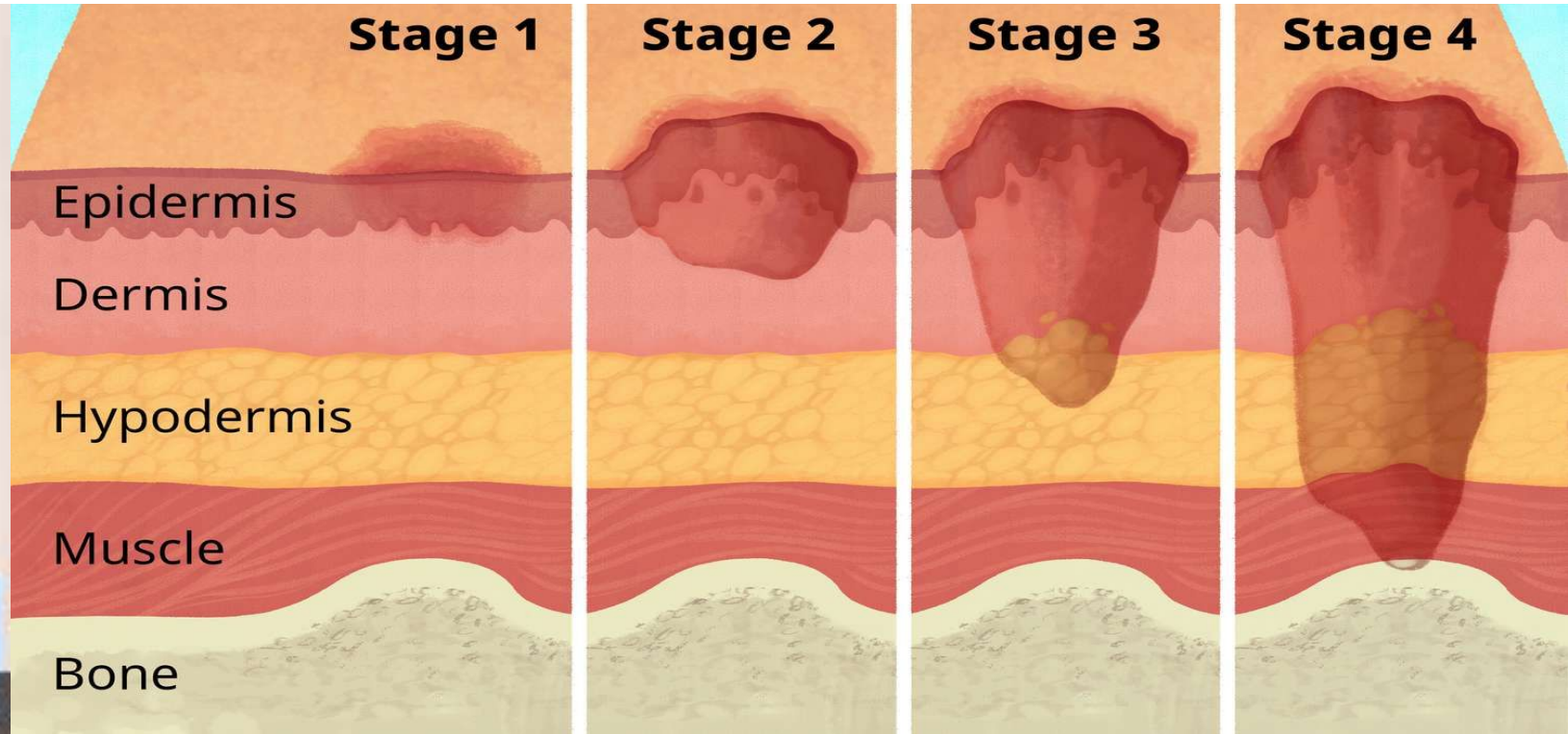




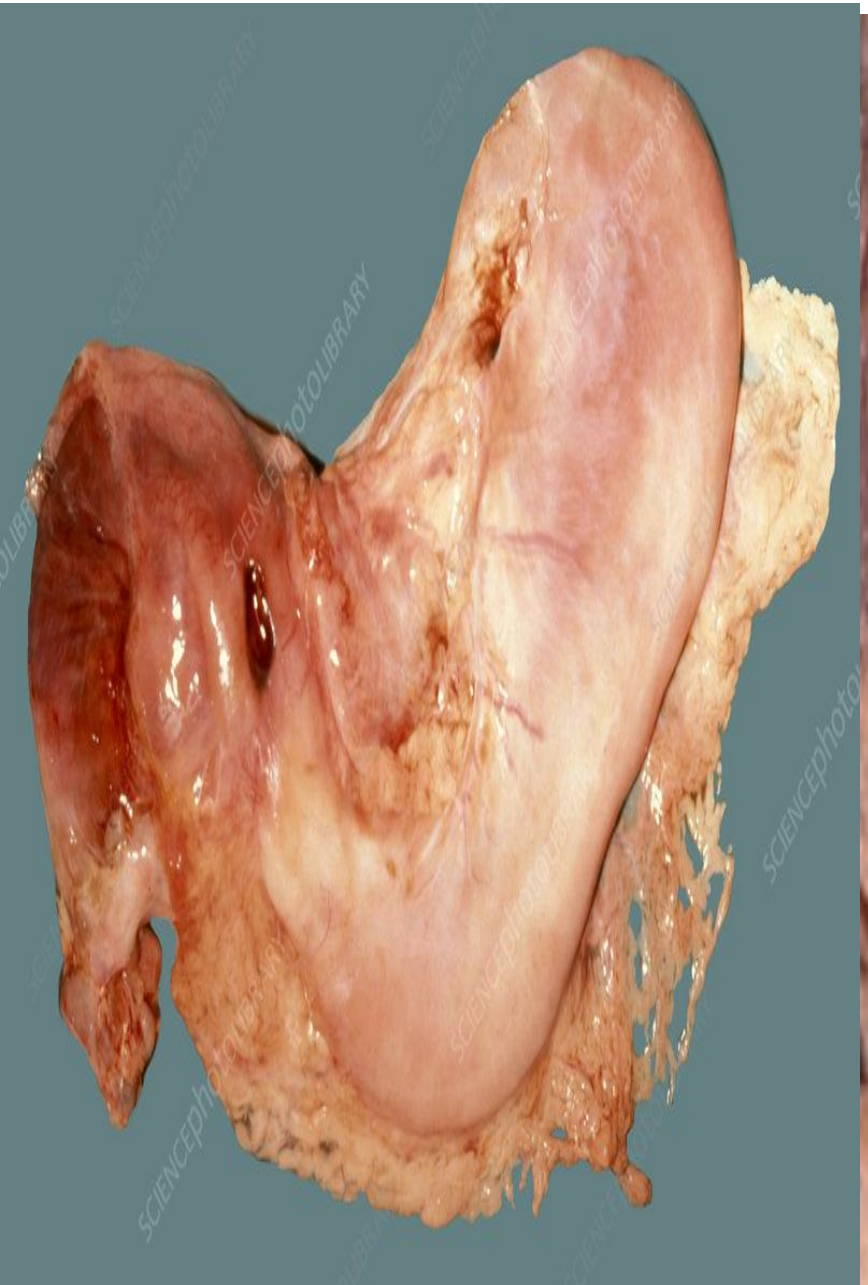
Ulcers

- Ulcerative inflammation refers to inflammation that is associated with a loss of epithelial lining that may extend into the deeper connective tissues
- Seen in mucosa of the mouth, stomach, intestines, or genital tract, skin and subcutaneous tissue





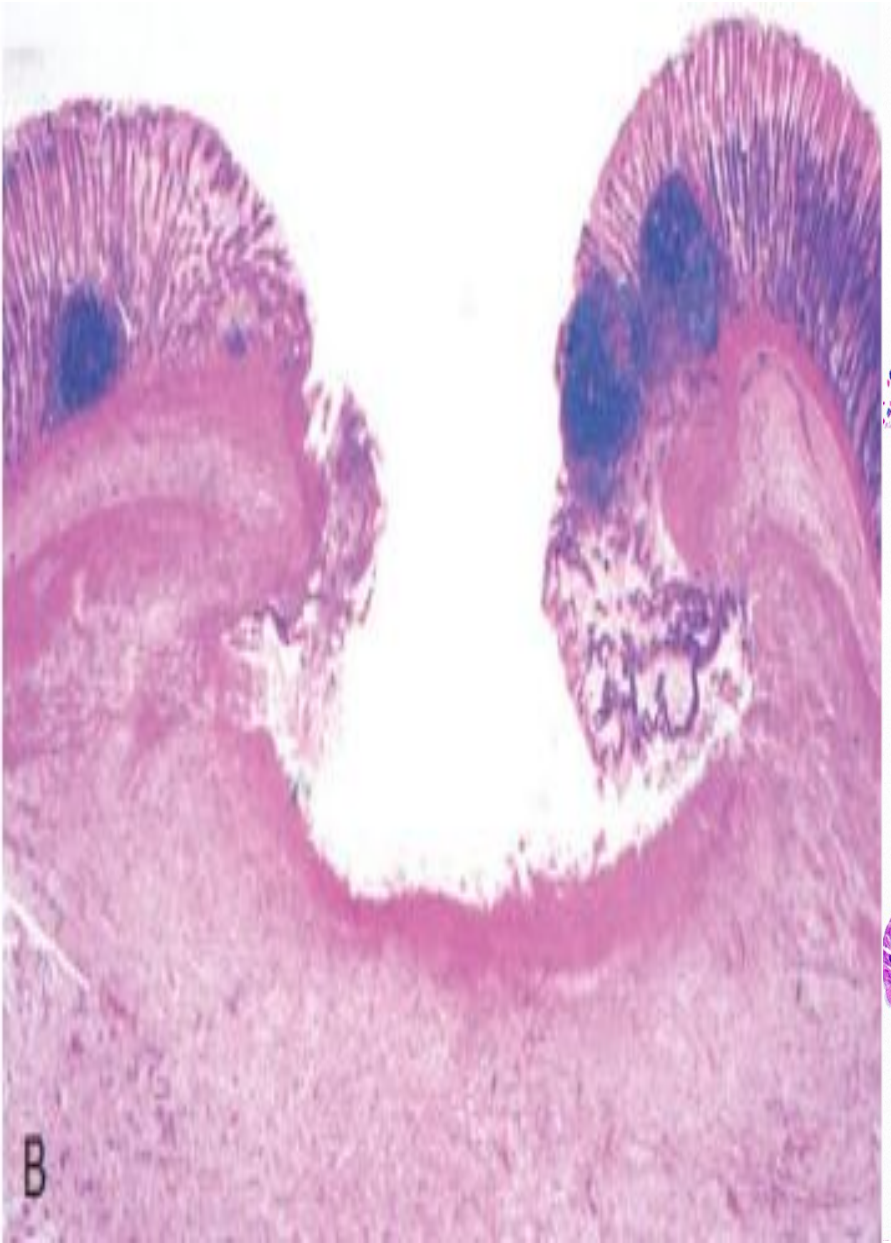
Ulcers

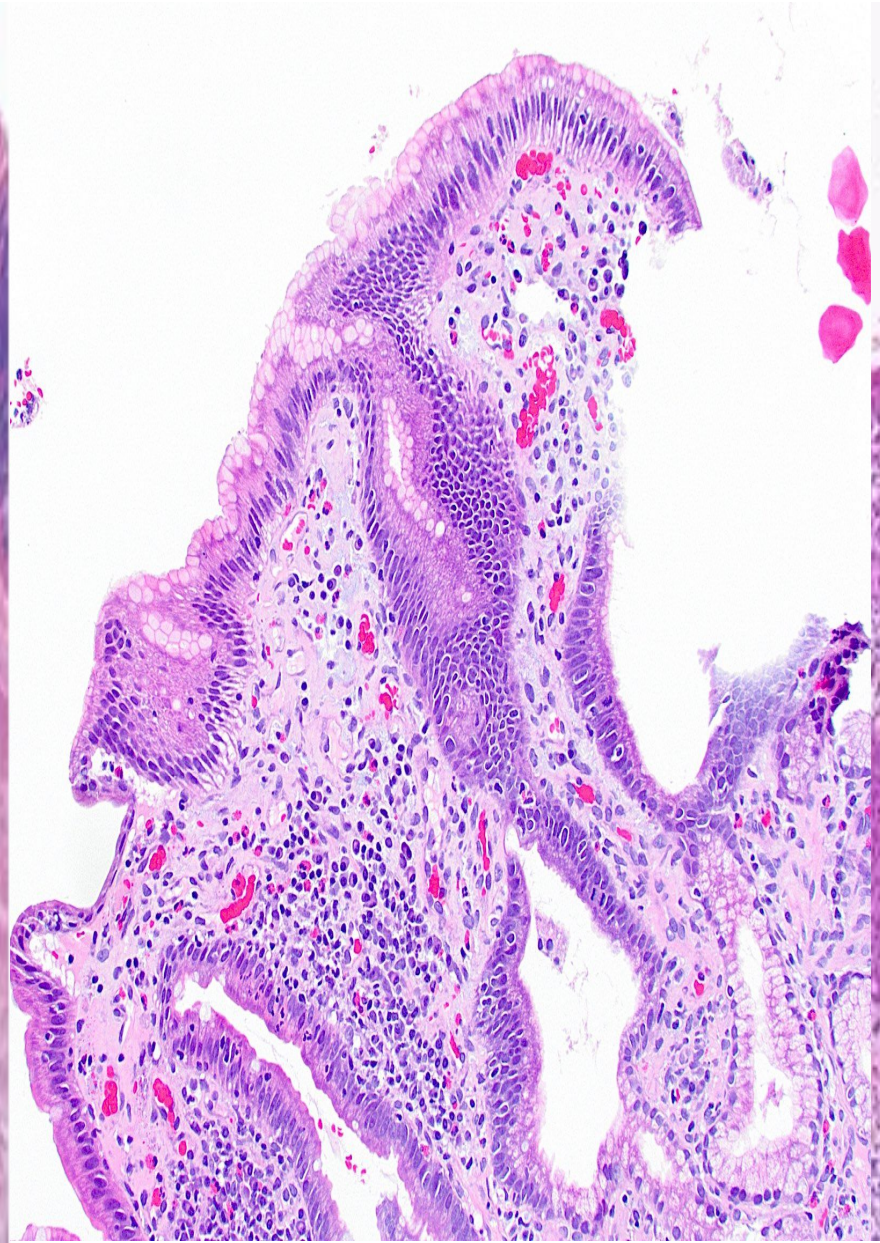


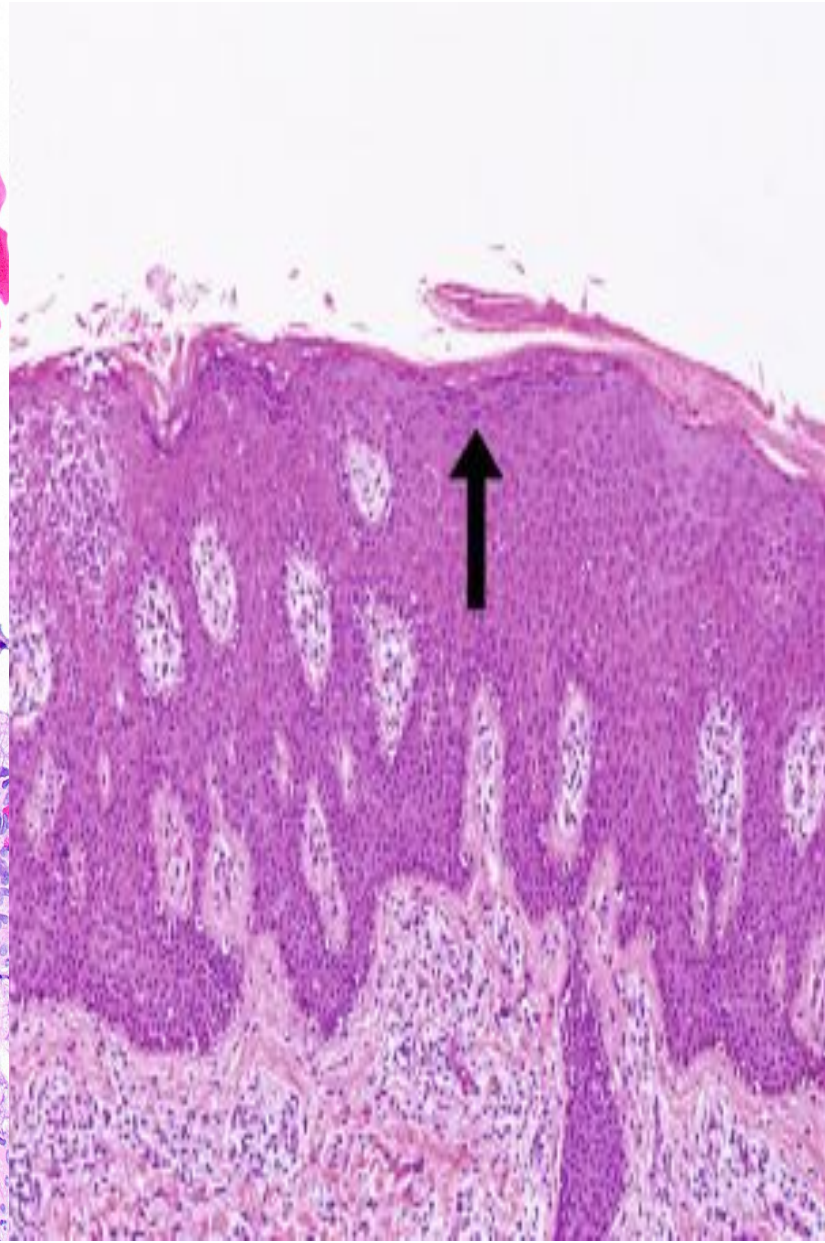




Ulcers

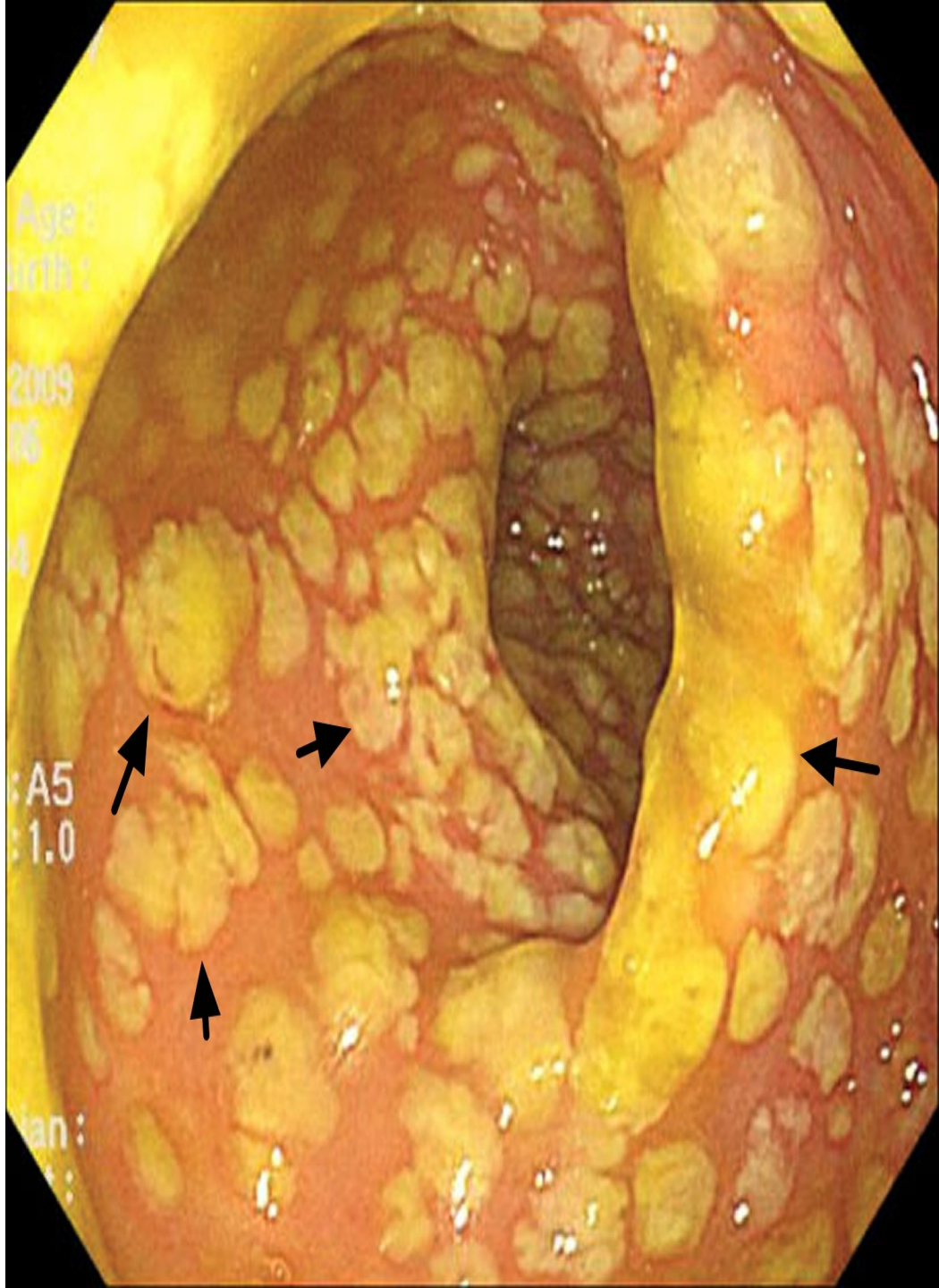






Pseudomembranous Inflammation

- Severe mucosal injury may be associated with extensive epithelial necrosis with sloughing. This creates large shallow ulcers
- Fibrin, dead epithelium, neutrophils, red cells and bacteria mix together to produce a white or cream-colored false (pseudo-) membrane covering the affected mucosa
- Diphtheria (Bordetella Pertussis) and pseudomembranous colitis (Clostridium difficile) are typical examples



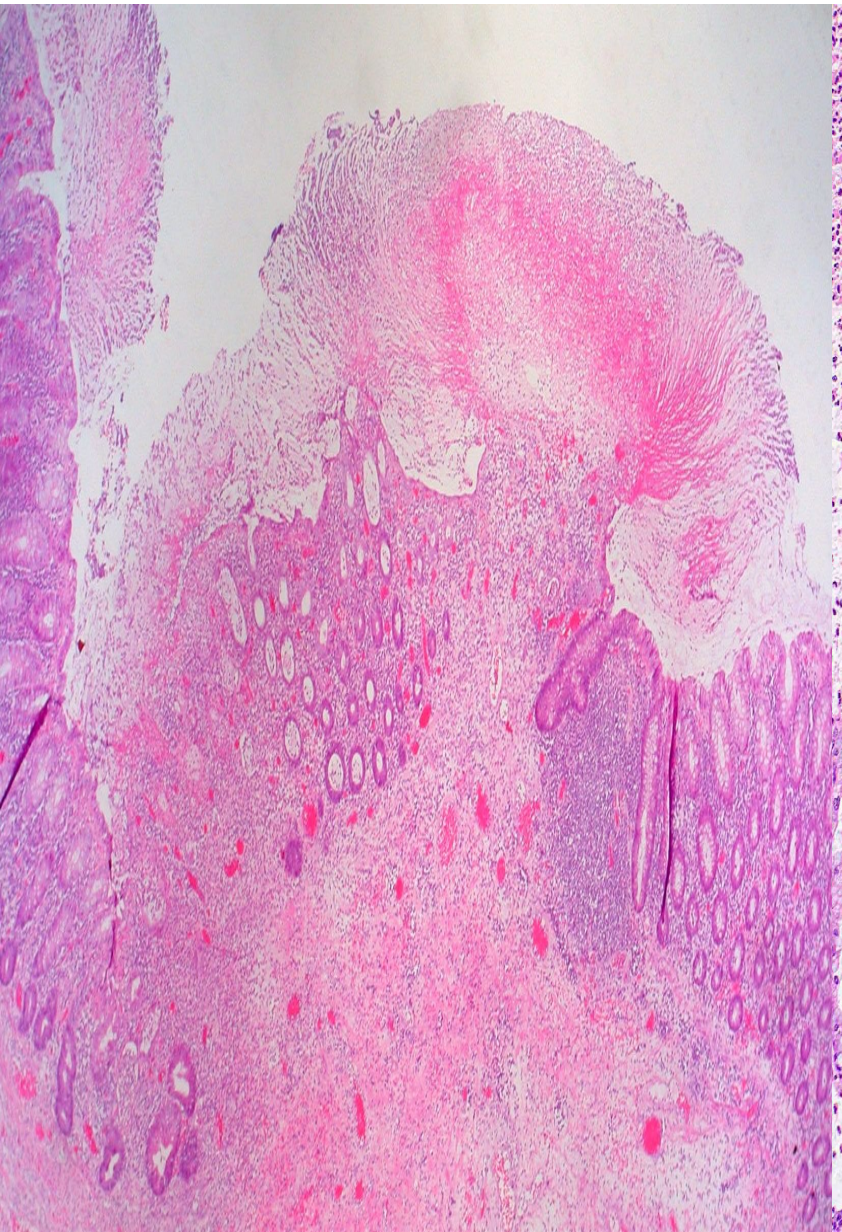
Pseudomembranous Inflammation

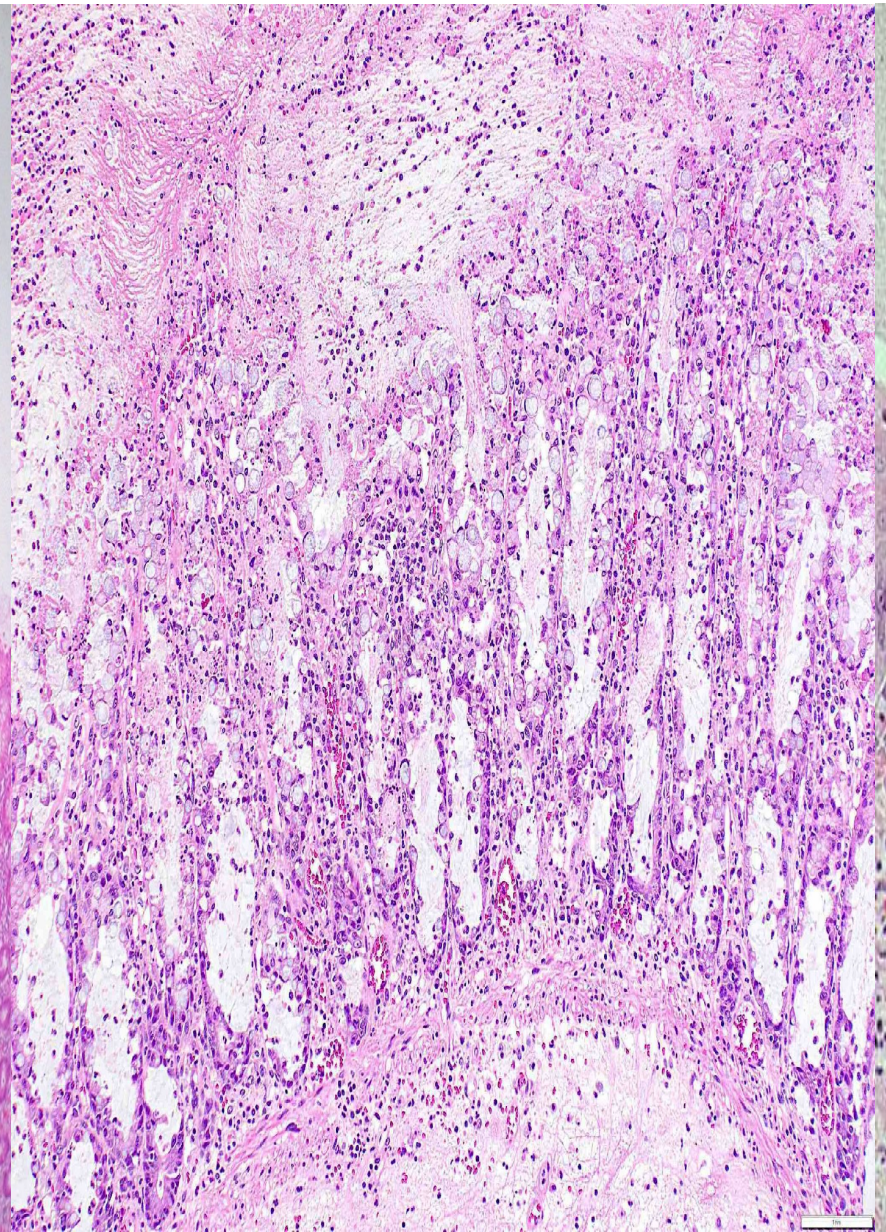


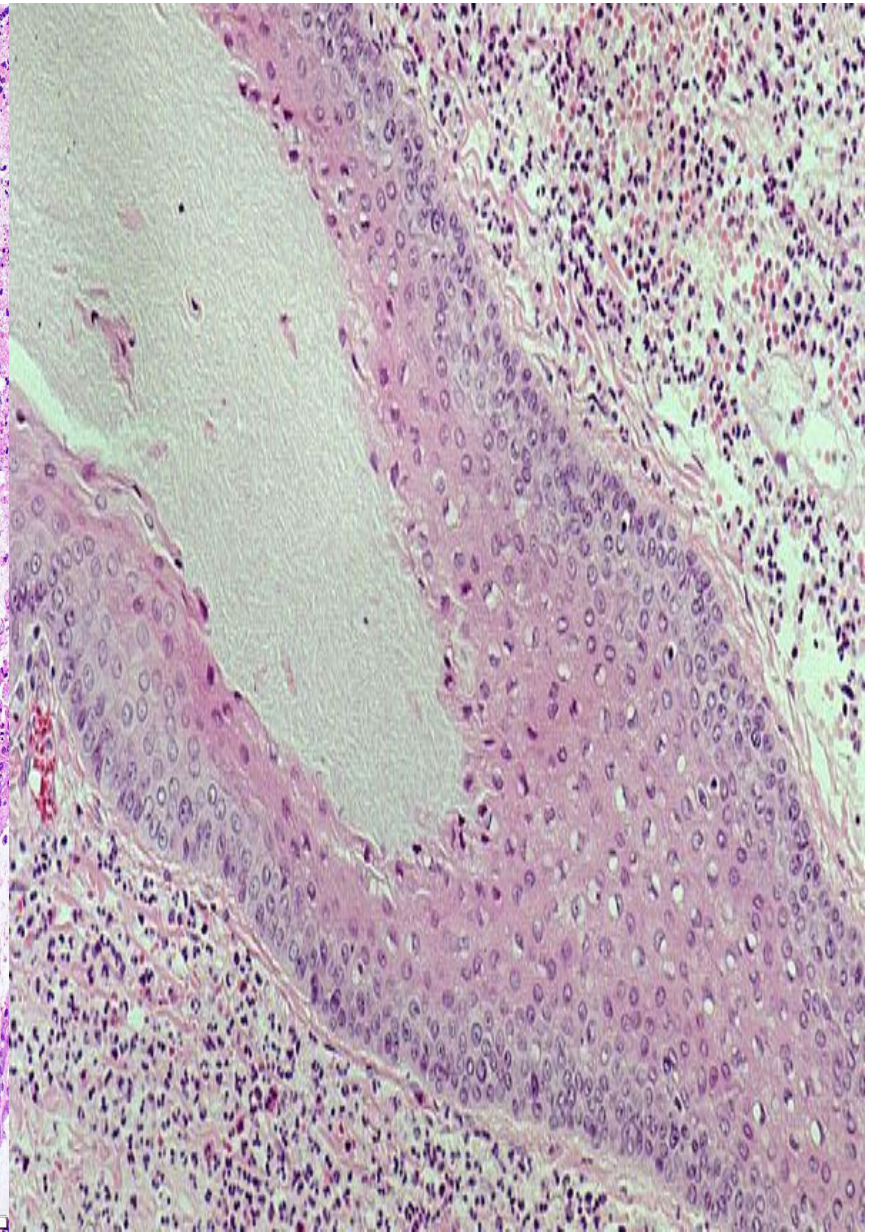




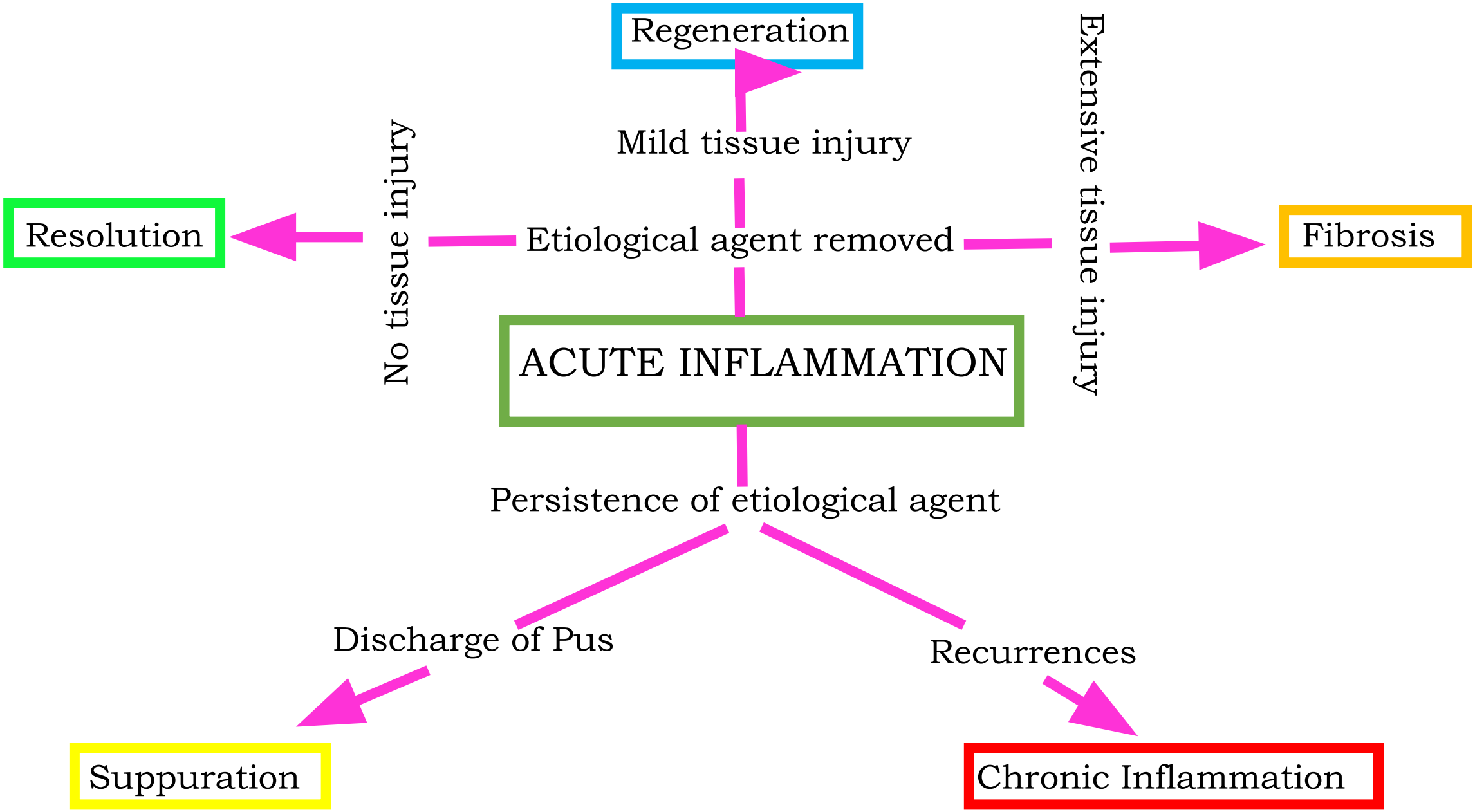
Pseudomembranous Inflammation



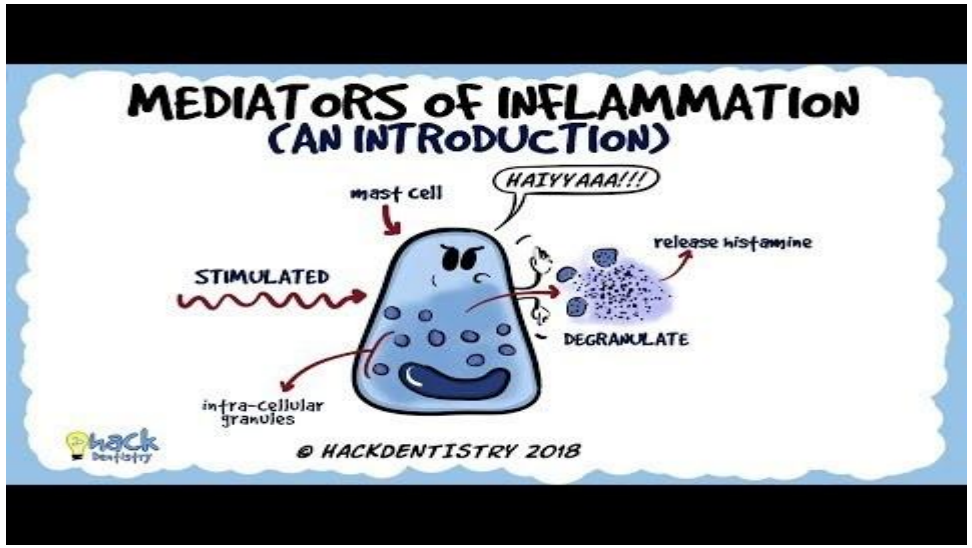


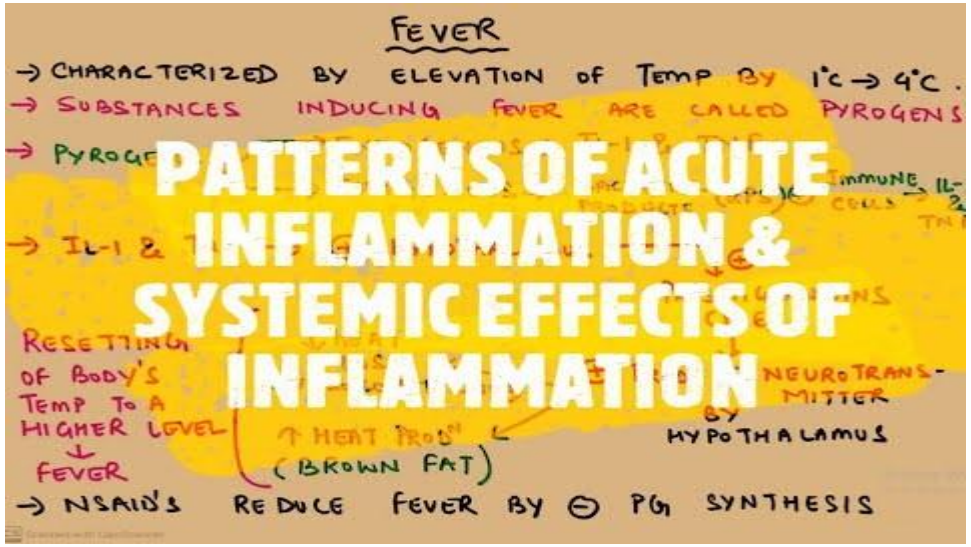


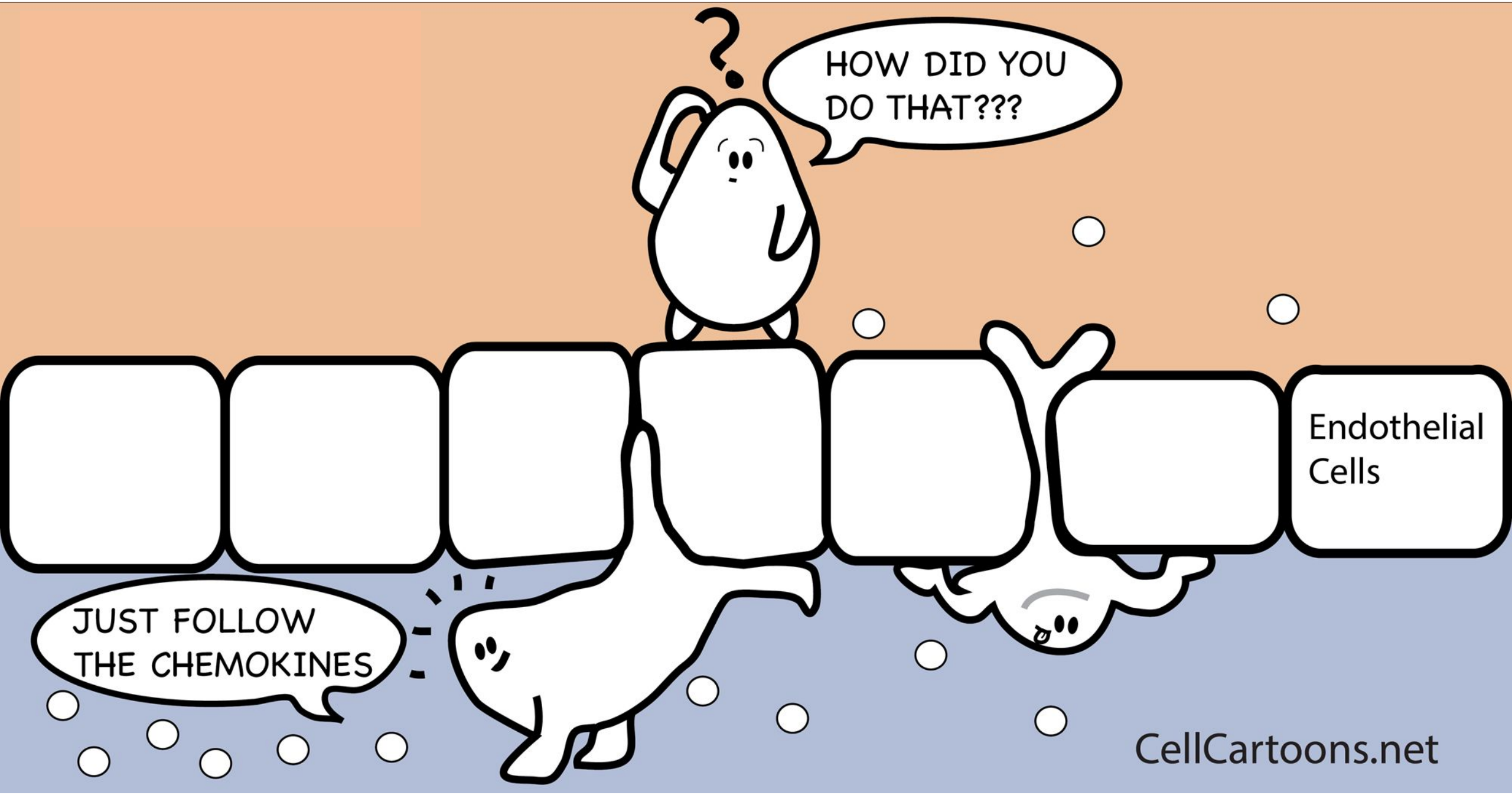
Outcome of Acute Inflammation



Additional Read






HOW DID YOU DO THAT???
Endothelial Cells
JUST FOLLOW THE CHEMOKINES
CellCartoons.net

Which of the following is a characteristic feature of fibrinous inflammation?

A) Purulent exudate
B) Fibrin-rich exudate
C) Serous fluid accumulation
D) Granuloma formation

A 50-year-old patient presents with a burn injury, and histological examination shows a clear, protein-poor fluid accumulation. This is an example of which type of acute inflammation?

A) Purulent
B) Fibrinous
C) Serous
D) Ulcerative

Pseudomembranous inflammation is most commonly seen in infections caused by:

A) Staphylococcus aureus
B) Clostridium difficile
C) Mycobacterium tuberculosis
D) Helicobacter pylori

Which of the following types of acute inflammation is characterized by pus formation?

A) Catarrhal
B) Fibrinous
C) Purulent
D) Serous

The key morphological feature of ulcerative inflammation is:

A) Presence of caseous necrosis
B) Superficial epithelial loss with inflammatory base
C) Abscess formation
D) Granulation tissue proliferation

Which of the following is a pre-formed mediator of inflammation?

A) Prostaglandins
B) Histamine
C) Leukotrienes
D) Bradykinin

The cyclooxygenase (COX) pathway is responsible for the synthesis of:

A) Leukotrienes
B) Histamine
C) Prostaglandins
D) Platelet-activating factor

Tumor Necrosis Factor-alpha (TNF-α) plays a crucial role in:

A) Vasodilation and permeability
B) Neutrophil chemotaxis
C) Fever
D) Platelet aggregation

Which inflammatory mediator is most potent in causing vasodilation and increasing vascular permeability?

A) Histamine
B) Interleukin-1
C) Platelet-activating factor
D) Interferon-gamma

A 21-year-old male presents to the dental OPD with pain and swelling in the lower jaw for the past 2 days. He had a decayed mandibular molar which he ignored. Clinical examination reveals a tender, fluctuant swelling near the angle of the mandible, with redness, warmth, and limited mouth opening. The dentist suspects a spreading odontogenic infection with acute inflammation.

Q1. What is the first vascular change in acute inflammation?
A1. Transient vasoconstriction followed by Vasodilation
Q2. Which cell type arrives first at the site of acute inflammation?
A2. Neutrophils
Q3. Name the key chemical mediator responsible for vasodilation.
A3. Histamine
Q4. Which mediators are responsible for pain in inflammation?
A4. Bradykinin, Prostaglandins
Q5. What arethe main mediators responsible for fever?
A5. Prostaglandins, Cytokines (IL-1,IL-6,TNF)
Q6. Which prostaglandins are responsible for vasodilation and fever?
A6. PGE2, PGD2, PGI2

Q7. What is the morphological term for pus formation?
A7. Suppuration/Purulent inflammation
Q8. What is the predominant pattern of inflammation in abscess?
A8. Suppurative/Purulent
Q9. Which complement fragments are powerful chemotactic agent?
A9. C3a, C5a
Q10. Q2. Common causative organism in pseudomembranous colitis?
A10. Clostridium
Q11. Appearance of pericardium in fibrinous inflammation?
A11. Bread-and-butter


NOT SURE IF I DID A GREAT JOB
OR NO ONE PAID ATTENTION
VIA 9GAG.COM

Mediator	Source	Action
Histamine (Vasoactive amines)	Mast cells, basophils, platelets	Vasodilation, increased vascular permeability, endothelial activation
Prostaglandins (Lipid products)	Mast cells, leukocytes	Vasodilation, pain, fever
Leukotrienes (Lipid products)	Mast cells, leukocytes	Increased vascular permeability, chemotaxis, leukocyte adhesion, and activation
Cytokines (TNF, IL-1, IL-6)	Macrophages, endothelial cells, mast cells	Local: endothelial activation (expression of adhesion molecules). Systemic: fever, metabolic abnormalities, hypotension (shock)
Chemokines	Leukocytes, activated macrophages	Chemotaxis, leukocyte activation
Platelet-activating factor	Leukocytes, mast cells	Vasodilation, increased vascular permeability, leukocyte adhesion, chemotaxis, degranulation, oxidative burst
Complement	Plasma (produced in liver)	Leukocyte chemotaxis and activation, direct target killing (membrane attack complex), vasodilation (mast cell stimulation)
Kinins	Plasma (produced in liver)	Increased vascular permeability, smooth muscle contraction, vasodilation, pain

Table 3-5 Principal Actions of Arachidonic Acid Metabolites in Inflammation

Action	Eicosanoid
Vasodilation	Prostaglandins PGI_2 (prostacyclin), PGE_1, PGE_2, PGD_2
Vasoconstriction	Thromboxane A_2, leukotrienes C_4, D_4, E_4
Increased vascular permeability	Leukotrienes C_4, D_4, E_4
Chemotaxis, leukocyte adhesion	Leukotrienes B_4, HETE

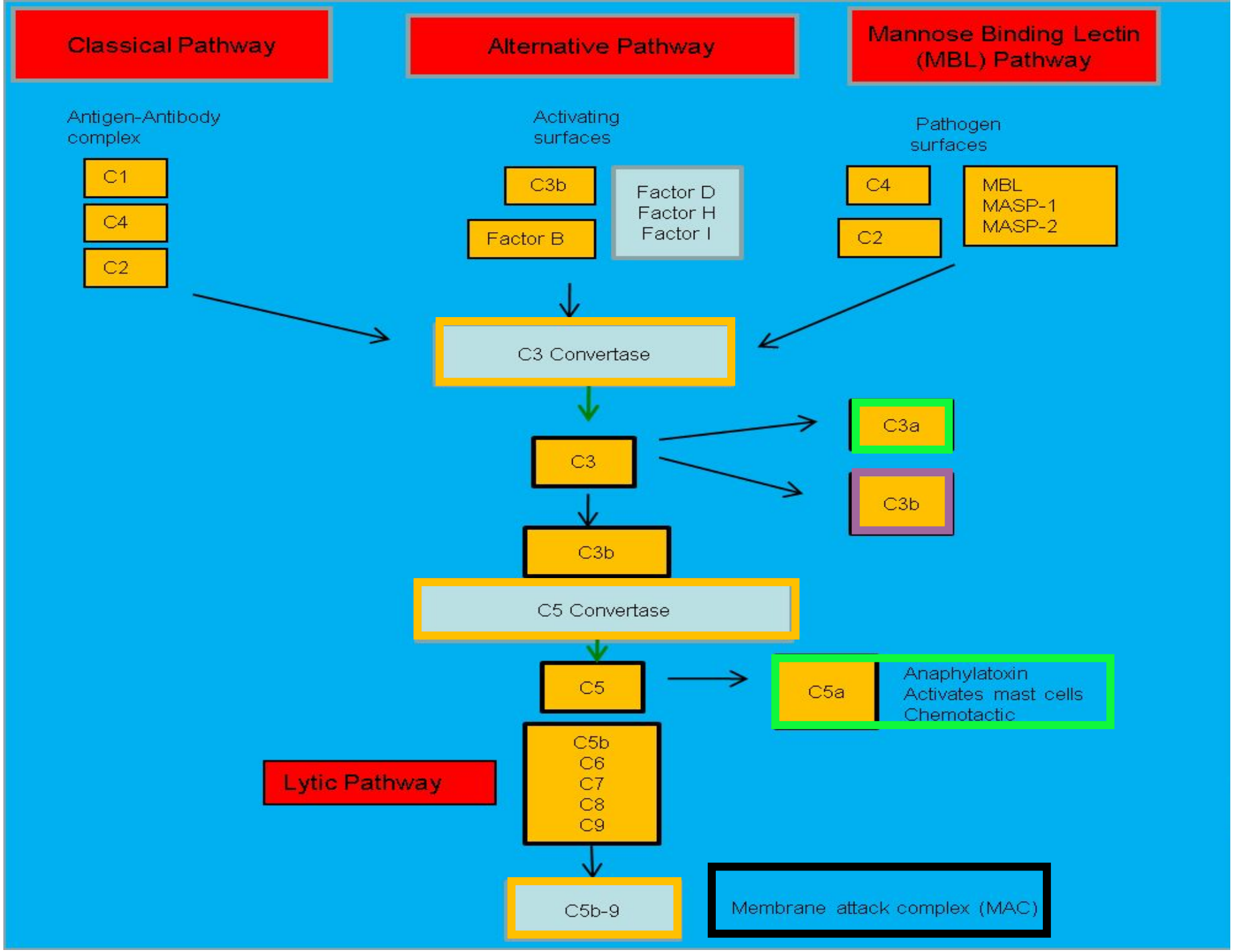
HETE, Hydroxyeicosatetraenoic acid.

Table 3-7 Role of Mediators in Different Reactions of Inflammation

Reaction of Inflammation	Principal Mediators
Vasodilation	Histamine Prostaglandins
Increased vascular permeability	Histamine and serotonin C3a and C5a (by liberating vasoactive amines from mast cells, other cells) Leukotrienes C_4, D_4, E_4
Chemotaxis, leukocyte recruitment and activation	TNF, IL-1 Chemokines C3a, C5a Leukotriene B_4
Fever	IL-1, TNF Prostaglandins
Pain	Prostaglandins Bradykinin
Tissue damage	Lysosomal enzymes of leukocytes Reactive oxygen species

Table 3-6 Cytokines in Inflammation

Cytokine	Principal Sources	Principal Actions in Inflammation
In Acute Inflammation		
TNF	Macrophages, mast cells, T lymphocytes	Stimulates expression of endothelial adhesion molecules and secretion of other cytokines; systemic effects
IL-1	Macrophages, endothelial cells, some epithelial cells	Similar to TNF; greater role in fever
IL-6	Macrophages, other cells	Systemic effects (acute phase response)
Chemokines	Macrophages, endothelial cells, T lymphocytes, mast cells, other cell types	Recruitment of leukocytes to sites of inflammation; migration of cells in normal tissues
IL-17	T lymphocytes	Recruitment of neutrophils and monocytes
In Chronic Inflammation		
IL-12	Dendritic cells, macrophages	Increased production of IFN-γ
IFN-γ	T lymphocytes, NK cells	Activation of macrophages (increased ability to kill microbes and tumor cells)
IL-17	T lymphocytes	Recruitment of neutrophils and monocytes


Classical Pathway
Alternative Pathway
Mannose Binding Lectin (MBL) Pathway
Antigen-Antibody complex
Activating surfaces
Pathogen surfaces
C1
C4
C2
C3b
Factor B
Factor D
Factor H
Factor I
C4
C2
MBL
MASP-1
MASP-2
C3 Convertase
C3
C3a
C3b
C3b
C5 Convertase
C5
C5a
Anaphylatoxin
Activates mast cells
Chemotactic
Lytic Pathway
C5b
C6
C7
C8
C9
C5b-9
Membrane attack complex (MAC)